

A1:DATA DASHBOARD

In preparation for a dashboard that is operational and supports executive decision-making, I have created a story that comprises two dashboards. The regional analysis dashboard and the sales analysis dashboard.

The Region Analysis Dashboard provides insights into customer retention and sales performance across different regions. For the regional analysis, I have created the following visualizations.

1. Stacked Bar Chart: Churn Count by Contract Type

The visual shows the number of customers per region by contract type, colored by the customers who have been retained and the customers who have churned

2. Side-by-Side Bar Chart: Region Retention and Churn Rates

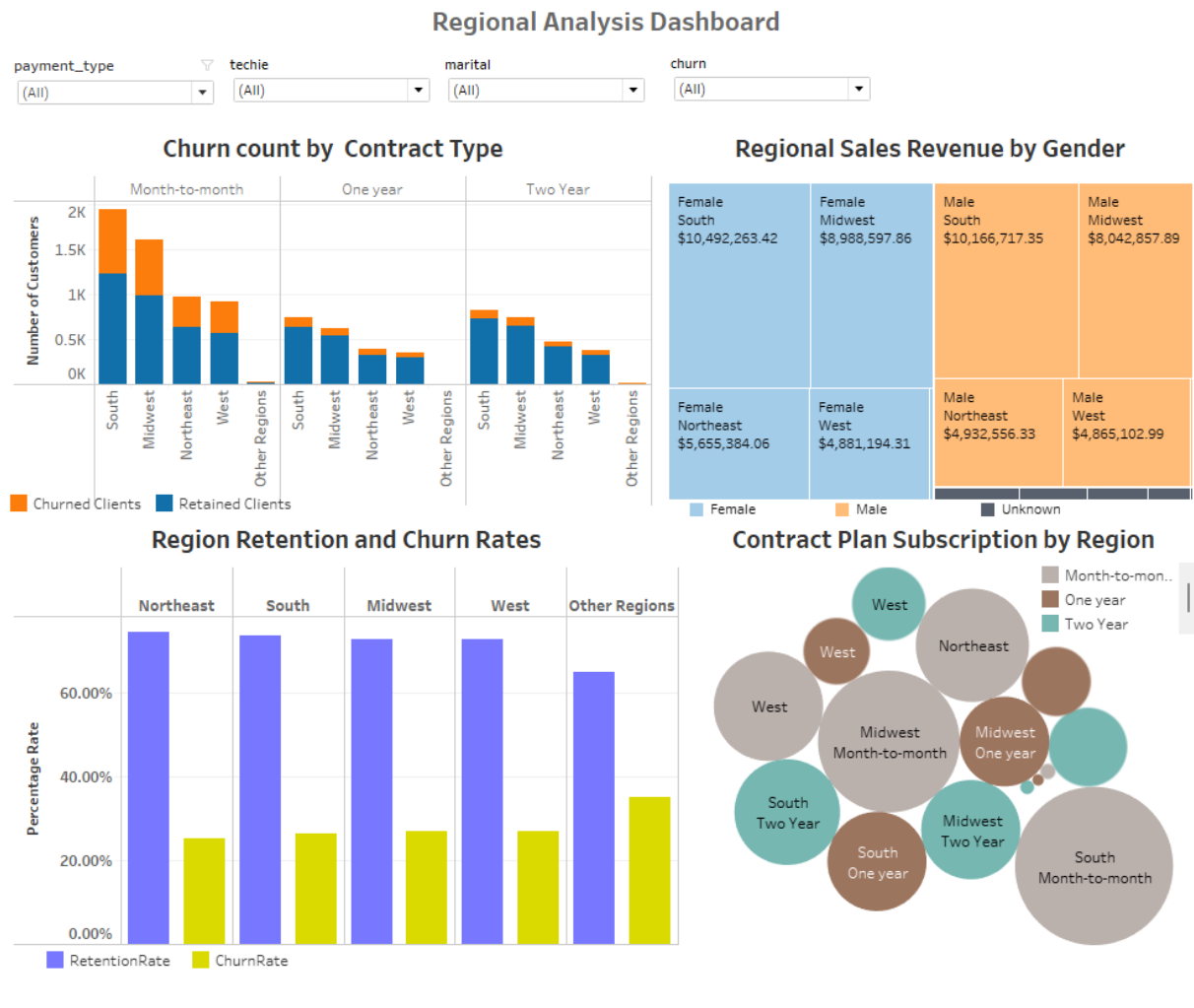
The visual compares churn and retention rates by region, with a tooltip for churn and retention rates by state.

3. Heatmap: Regional Sales Revenue by Gender

The visual displays sales revenue by gender with regional insights, with a tooltip for sales and monthly charges by state.

4. Bubble Chart: Contract Plan Subscription by Region

The visual represents contract subscriptions by region. With a tooltip that shows the customer count, sales amount, and average monthly charge



The Sales Dashboard provides key insights into customer behavior and sales performance. For the Sales analysis, I have created the following visualizations

1. Stacked Bar Chart: Monthly Charges by State

The visual shows the Monthly charge by state, colored by the customers who have been retained and the customers who have churned, with a tooltip that shows the monthly charges and average monthly charge.

2. Heatmap: Regional Sales Revenue by Gender

The visual displays sales revenue by gender with regional insights, with a tooltip for sales and monthly charges by state.

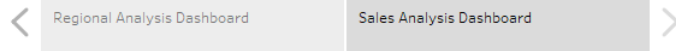
3. Map: Sales revenue by state

The visual displays the total sales revenues that have been collected from each state. The total sales revenue is computed by multiplying the monthly charges by the tenure of the customer.

4. Bubble Chart: Sales by Contract Type

The visual represents sales by contract subscription with a tooltip showing the number of customers, the average monthly charge for the contract subscription, and the total monthly charges recovered for the customer.

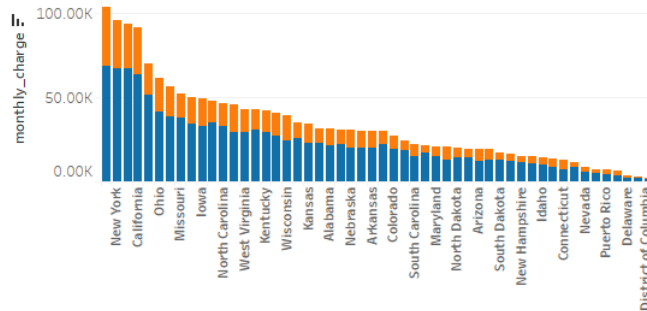
Telecommunication Churn Data Analysis



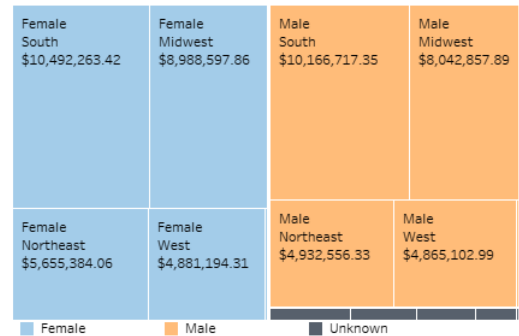
Sales Analysis Dashboard

payment_type: (All) | techie: (All) | marital: (All) | churn: (All) | region: (All)

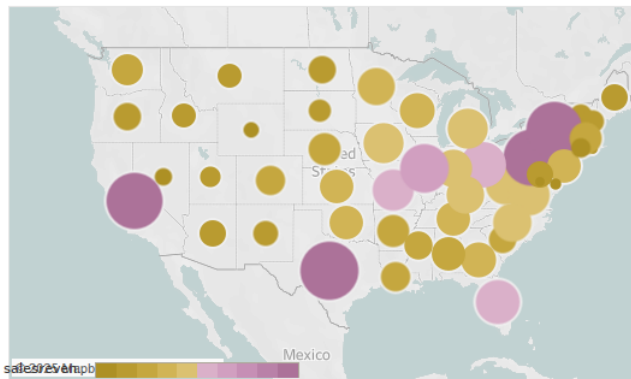
Monthly Charges by State



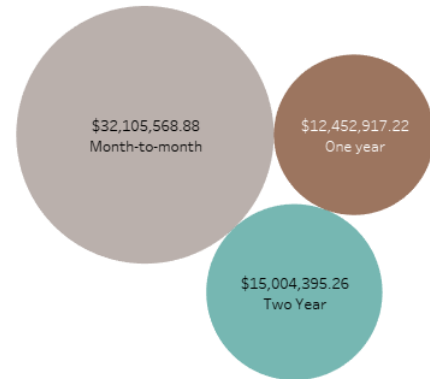
Regional Sales Revenue by Gender



Sales revenue by State



Sales by Contract Type



A1: DATASETS AND DASHBOARD FILE

1. The telecommunication industry Churn dataset under the Churn database
2. External data from Kaggle USA States and regions (states.csv). The dataset has 51 rows and 4 features. The data has been loaded into a database table called states in the Churn dataset

Churn database

D211: AdvancedDataAcquisition_TableauDataDashboard

BeatriceM_WGU_MSDA5

Connections

localhost
PostgreSQL

Database

churn

Table

contract

customer

job

location

payment

states

New Custom SQL

New Union

New Table Extension

churn/postgres@PostgreSQL 16															
Data Output Messages															
	customer_id	marital	churn	gender	salesrevenue	monthly_charge	techie	Contract type	city	state	county	payment_type	state_name	region	
1	K409198	Widowed	Retained Clients	Male	1171.92	172.46	No TechieSupport	One year	Calhoun	IL	Richland	Credit Card Automatic	Illinois	Midwest	
2	S120509	Married	Churned Clients	Female	280.65	242.63	Has Techie Support	Month-to-month	Neptune Beach	FL	Duval	Bank Transfer Automatic	Florida	South	
3	K191035	Widowed	Retained Clients	Female	2519.84	159.95	Has Techie Support	Two Year	Linesville	PA	Crawford	Credit Card Automatic	Pennsylvania	Northeast	
4	D90850	Married	Retained Clients	Male	2049.73	119.96	Has Techie Support	Two Year	Dawson	ND	Kidder	Mailed Check	North Dakota	Midwest	
5	K662701	Separated	Churned Clients	Male	250.56	149.95	No TechieSupport	Month-to-month	Sabana Seca	PR	Toa Baja	Mailed Check	Puerto Rico	Other Regions	
6	W303516	Never Married	Retained Clients	Female	1295.24	185.01	No TechieSupport	One year	London	KY	Laurel	Electronic Check	Kentucky	South	
7	U335188	Widowed	Churned Clients	Male	2648.92	200.12	Has Techie Support	Month-to-month	Burton	MI	Genesee	Electronic Check	Michigan	Midwest	
8	V538685	Married	Churned Clients	Female	490.18	114.95	Has Techie Support	Month-to-month	Whiteface	TX	Cochran	Mailed Check	Texas	South	
9	M716771	Separated	Retained Clients	Unknown	965.67	117.47	No TechieSupport	Month-to-month	New Hampton	NH	Belknap	Bank Transfer Automatic	New Hampshi...	Northeast	
10	I676080	Married	Retained Clients	Female	556.03	162.48	No TechieSupport	Two Year	Ottertail	MN	Otter Tail	Mailed Check	Minnesota	Midwest	
11	J980369	Separated	Retained Clients	Female	3370.96	174.96	No TechieSupport	Month-to-month	Mc Lemosresville	TN	Carroll	Mailed Check	Tennessee	South	
12	E243720	Married	Retained Clients	Female	1577.90	149.96	No TechieSupport	Month-to-month	Dix	IL	Jefferson	Electronic Check	Illinois	Midwest	
13	F139569	Divorced	Retained Clients	Male	1788.29	137.44	No TechieSupport	Month-to-month	Chidester	AR	Ouachita	Bank Transfer Automatic	Arkansas	South	
14	X44200	Separated	Retained Clients	Female	3122.17	184.97	No TechieSupport	Two Year	West Lebanon	NY	Columbia	Bank Transfer Automatic	New York	Northeast	
15	H68068	Separated	Churned Clients	Male	1609.29	159.97	No TechieSupport	Month-to-month	Sabina	OH	Clinton	Bank Transfer Automatic	Ohio	Midwest	
16	A403906	Divorced	Churned Clients	Male	2464.02	177.65	No TechieSupport	One year	Smyer	TX	Hockley	Electronic Check	Texas	South	
17	V847470	Married	Churned Clients	Female	3076.99	194.97	No TechieSupport	Month-to-month	Fortuna	CA	Humboldt	Bank Transfer Automatic	California	West	
18	F721878	Separated	Churned Clients	Male	466.85	202.68	No TechieSupport	Two Year	Atlantic Mine	MI	Houghton	Mailed Check	Michigan	Midwest	
19	F487435	Widowed	Retained Clients	Female	2609.11	152.49	Has Techie Support	Two Year	Jamison	PA	Bucks	Bank Transfer Automatic	Pennsylvania	Northeast	
20	B561228	Divorced	Churned Clients	Male	1920.22	149.94	No TechieSupport	Month-to-month	Houston	TX	Harris	Credit Card Automatic	Texas	South	
21	X325271	Never Married	Retained Clients	Male	3783.53	184.98	No TechieSupport	Month-to-month	Bethlehem	KY	Henry	Bank Transfer Automatic	Kentucky	South	
22	I282896	Divorced	Retained Clients	Male	308.03	127.50	No TechieSupport	Two Year	Oak Park	IL	Cook	Bank Transfer Automatic	Illinois	Midwest	
23	I367206	Widowed	Retained Clients	Male	831.30	124.96	No TechieSupport	Month-to-month	Bonneau	SC	Berkeley	Electronic Check	South Carolina	South	
24	N37182	Divorced	Retained Clients	Male	1281.12	149.95	No TechieSupport	One year	Kansas City	KS	Wyandotte	Electronic Check	Kansas	Midwest	
25	U644626	Widowed	Churned Clients	Female	937.76	162.51	No TechieSupport	Month-to-month	Florissant	MO	St. Louis	Credit Card Automatic	Missouri	Midwest	
26	A766555	Separated	Churned Clients	Male	384.40	92.46	No TechieSupport	Month-to-month	Hoit	FL	Okaloosa	Bank Transfer Automatic	Florida	South	
27	V54032	Married	Churned Clients	Female	2316.82	222.65	No TechieSupport	Month-to-month	Coin	IA	Page	Mailed Check	Iowa	Midwest	
28	Z301326	Widowed	Churned Clients	Female	194.91	139.98	No TechieSupport	Month-to-month	Afton	TX	Dickens	Electronic Check	Texas	South	
29	J887250	Separated	Churned Clients	Female	286.27	200.13	Has Techie Support	One year	Yanell	AZ	Yavapai	Bank Transfer Automatic	Arizona	West	
30	Z686376	Never Married	Churned Clients	Male	1397.98	257.65	No TechieSupport	Two Year	Minneapolis	MN	Hennepin	Credit Card Automatic	Minnesota	Midwest	

D211: AdvancedDataAcquisition_TableauDataDashboard BeatriceM_WGU_MSDA6

D211ReportingQuery is made of 1 table. ⓘ

Custom SQL Query

Edit Custom SQL Query

Duplicate

D211ReportingQuery

18 fields 10000 rows

100 → rows ⓘ

Name

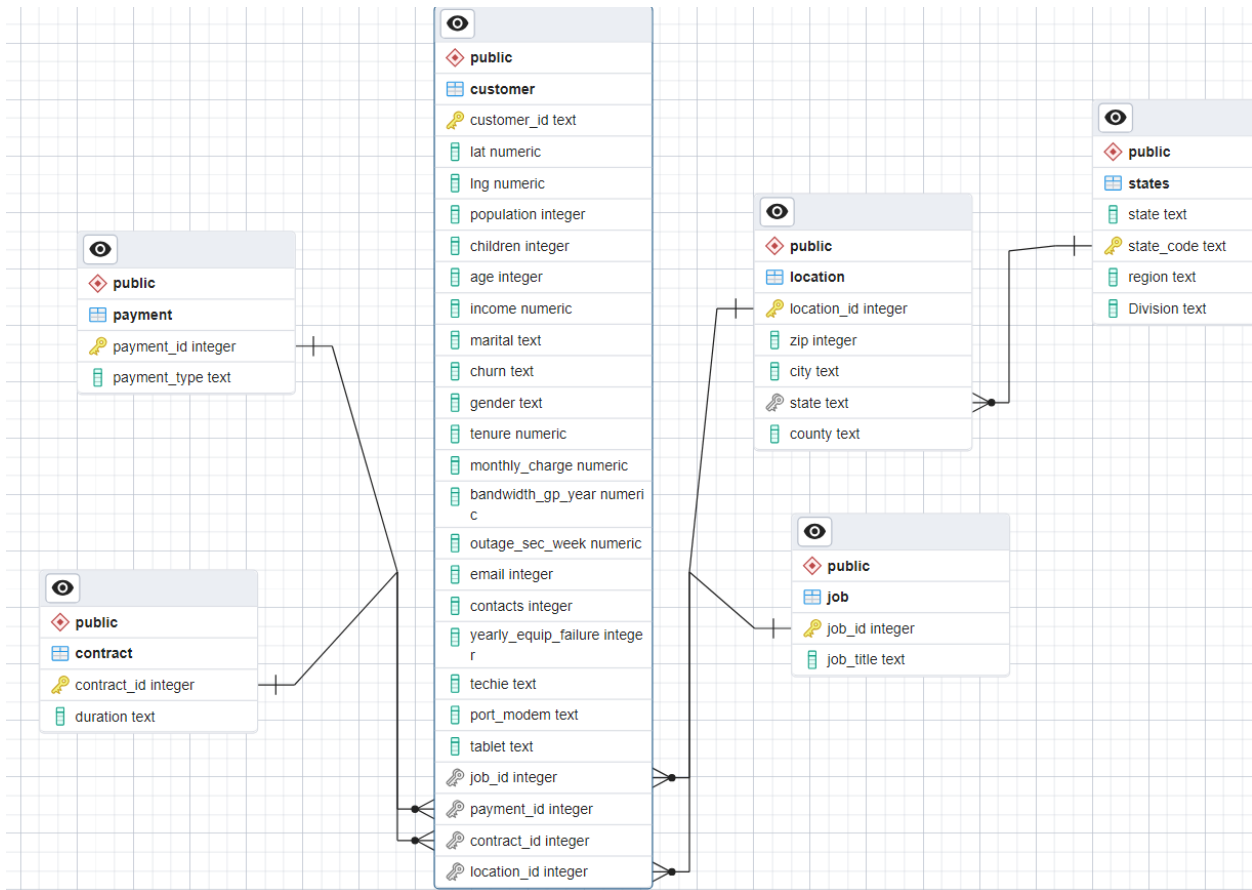
D211ReportingQuery

Fields

Type	Field Name	Physical Table	Remote...
Abc	customer_id	Custom SQL Query	customer...
Abc	marital	Custom SQL Query	marital
Abc	churn	Custom SQL Query	churn
Abc	gender	Custom SQL Query	gender
#	salesrevenue	Custom SQL Query	salesreve...
#	monthly_charge	Custom SQL Query	monthly_...
Abc	techie	Custom SQL Query	techie
Abc	Contract type	Custom SQL Query	Contract ...

Abc Custom SQL Query	Abc Custom SQL Query	Abc Custom SQL Query	Abc Custom SQL Query	# Custom SQL Query	# Custom SQL Query	Abc Custom SQL Query	Abc Custom SQL Query	Custom SQL Query	Custom SQL Q
customer_id	marital	churn	gender	salesrevenue	monthly_charge	techie	Contract type	city	state
K409198	Widowed	Retained Clients	Male	\$1171.92	\$172.46	No TechieSupport	One year	Calhoun	IL
S120509	Married	Churned Clients	Female	\$280.65	\$242.63	Has Techie Support	Month-to-month	Neptune Beach	FL
K191035	Widowed	Retained Clients	Female	\$2,519.84	\$159.95	Has Techie Support	Two Year	Linesville	PA
D90850	Married	Retained Clients	Male	\$2,049.73	\$119.96	Has Techie Support	Two Year	Dawson	ND
K662701	Separated	Churned Clients	Male	\$250.56	\$149.95	No TechieSupport	Month-to-month	Sabana Seca	PR
W303516	Never Married	Retained Clients	Female	\$1,295.24	\$185.01	No TechieSupport	One year	London	KY
U335188	Widowed	Churned Clients	Male	\$2,648.92	\$200.12	Has Techie Support	Month-to-month	Burton	MI
V538685	Married	Churned Clients	Female	\$490.18	\$114.95	Has Techie Support	Month-to-month	Whiteface	TX
M716771	Separated	Retained Clients	Unknown	\$965.67	\$117.47	No TechieSupport	Month-to-month	New Hampton	NH
I676080	Married	Retained Clients	Female	\$556.03	\$162.48	No TechieSupport	Two Year	Ottertail	MN
J980369	Separated	Retained Clients	Female	\$3,370.96	\$174.96	No TechieSupport	Month-to-month	Mc Lemoresv...	TN
E243720	Married	Retained Clients	Female	\$1,577.90	\$149.96	No TechieSupport	Month-to-month	Dix	IL

The ERD diagram for the database tables used for the Churn analysis



A2: DASHBOARD INSTALLATION

Step-by-Step Dashboard Installation Guide

1. Prerequisites

Before installing the dashboard, ensure you have the following:

- A computer with Windows or macOS
- Tableau Desktop or Tableau Reader installed
- Access to the dashboard file (.twbx or .twb)



Beatrice
Mwangi_Task1 _D21

2. Installing Tableau (If Not Already Installed)

- Visit the Tableau website (<https://www.tableau.com/>).
- Download Tableau Desktop (for full functionality) or Tableau Reader (for viewing only).
- Follow the on-screen instructions to install the software.
- Once installed, open Tableau.

3. Opening the Dashboard

- Locate the dashboard file (.twbx or .twb) on your computer.
- Double-click the file to open it in Tableau Desktop or Tableau Reader.

4. Connecting to Data (If required)

- The data is saved as an extract; however, if prompted, the user may need to connect to a database. Use the credentials below
- The prompt may also require the creation of the External dataset table and loading of the data (Table creation script, data file attached)



External data SQL
table.sql



states.csv

Import/Export data - table 'states'

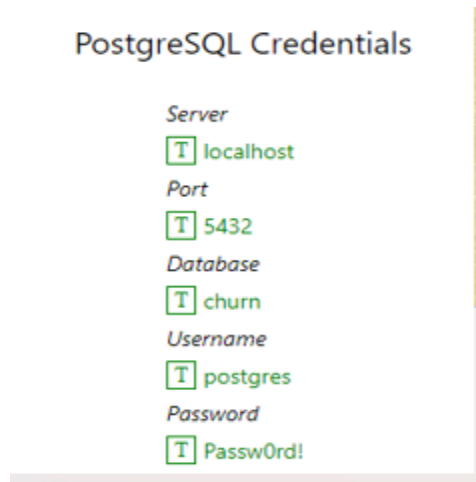
General Options Columns

Import/Export ☒ Import ☐ Export

Filename C:\Users\beatr\OneDrive\Desktop\WGU\D211\Externaldata\states.csv 

Format csv | v

Encoding UTF8 x | v



- Verify that the data loads correctly into Tableau.

5. Verifying Installation

- Ensure all visuals and filters are displayed correctly.
- Test interactions such as clicking on charts and applying filters.
- If any issues arise, confirm that data sources are correctly linked and try refreshing the data.

A3: DASHBOARD NAVIGATION

- **Overview**

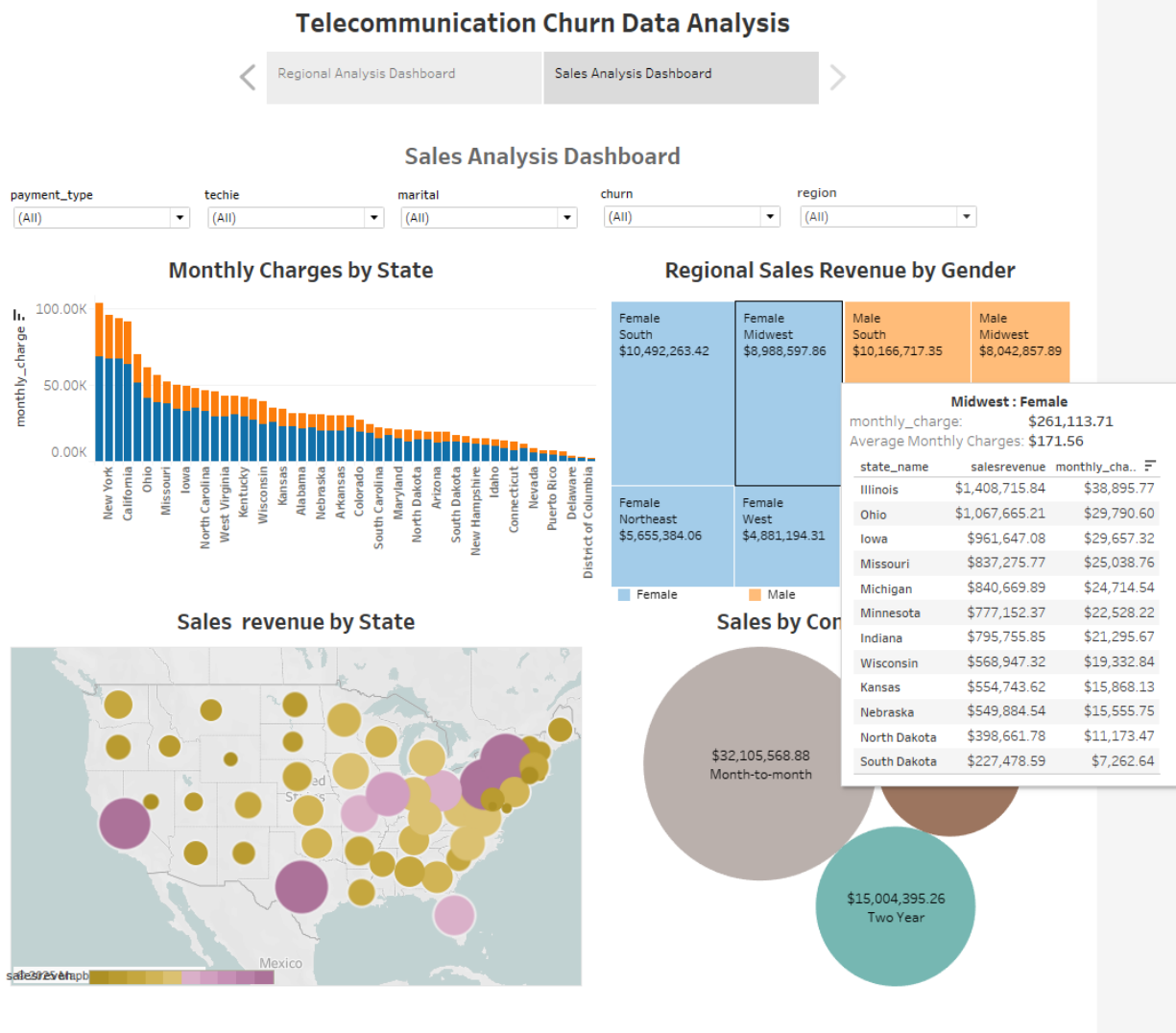
The telecommunication Industry Churn data analysis story is designed for intuitive and interactive analysis, helping executive leaders make data-driven decisions. Users can gain deep insights into customer behavior, sales trends, and regional performance by using the filtering options, exploring tooltips, and interacting with the visuals.

The interactive dashboard provides insights into customer engagement, sales performance, and regional trends via the regional analysis dashboard and Sales Analysis dashboard

- Navigating the Dashboards

1. Interacting with Visuals

Hover Over Charts: Hover your cursor over any chart or data point to view additional details in the tooltip. Tooltips provide insights such as total sales revenue, churn rates, and other key metrics.

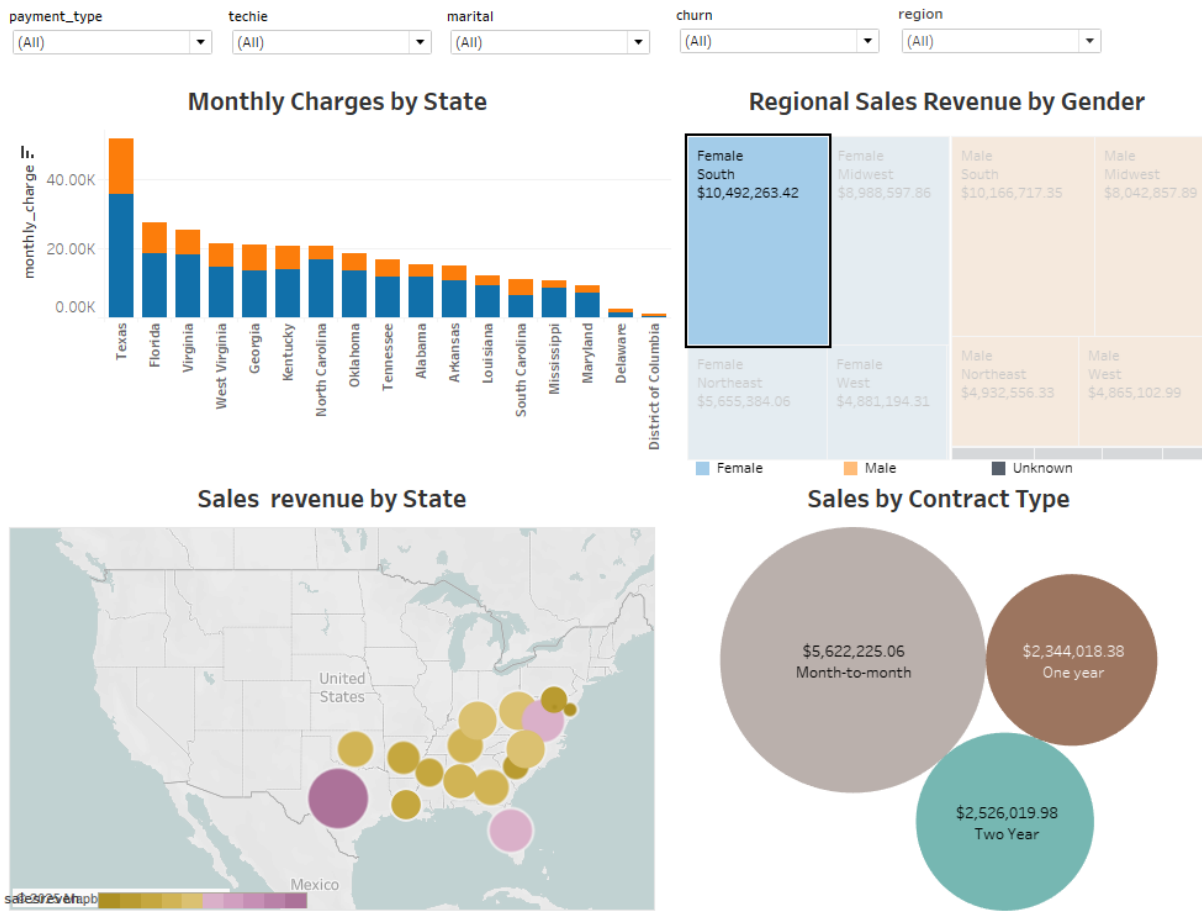


Click on Data Points: Clicking on elements within a chart (such as a specific region, contract type, or Gender) filters other visuals in the dashboard to reflect relevant data.

Telecommunication Churn Data Analysis



Sales Analysis Dashboard



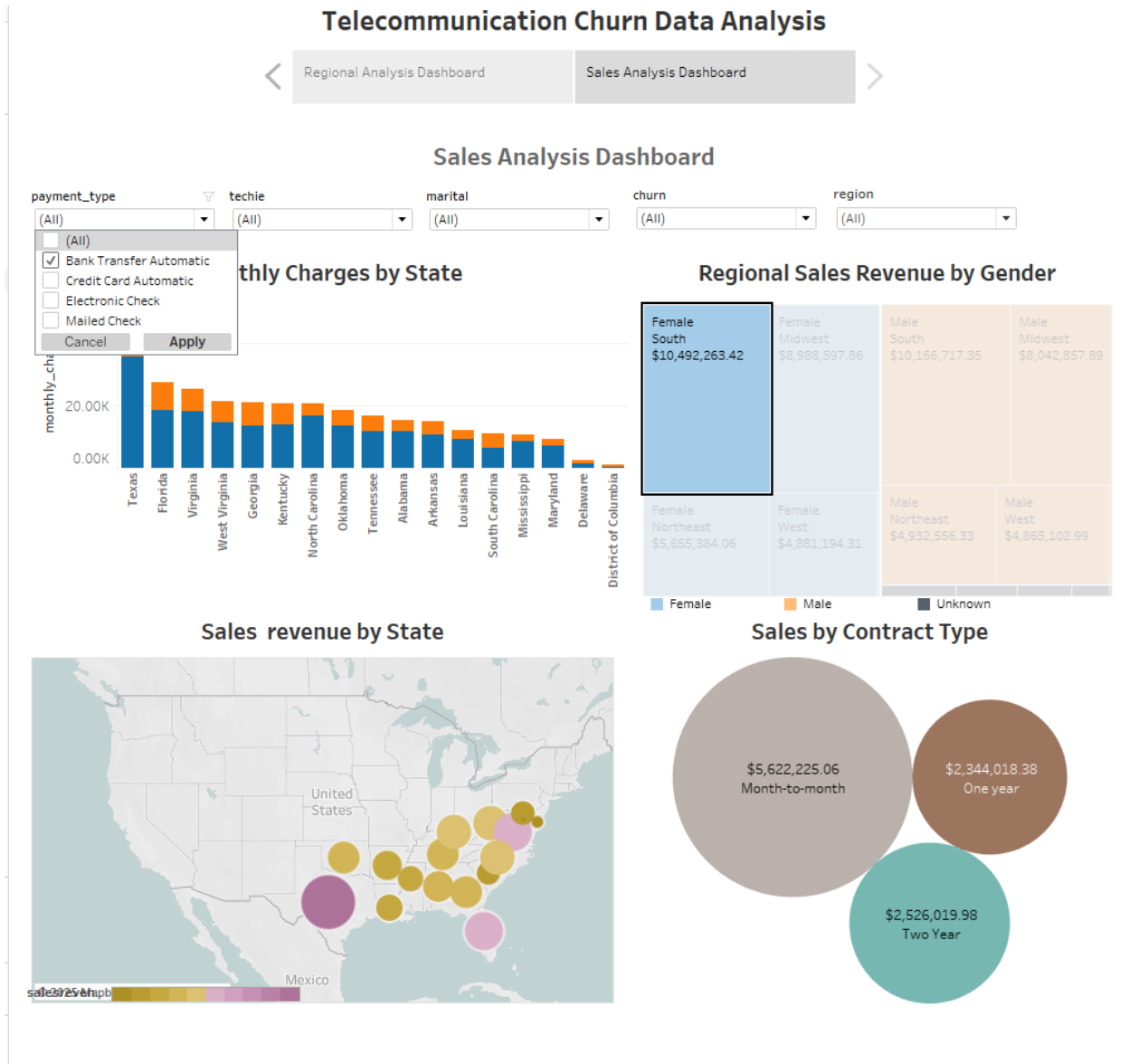
2. Using Filters

Each dashboard includes the filters Payment Method, Tech, Marital, and Churn that allow users to refine data views.

To apply filters:

- Locate the filter panel on the side of the dashboard.
- Click on dropdown menus to select specific categories

- Select one or multiple filter options to update the visuals dynamically.
- Click apply

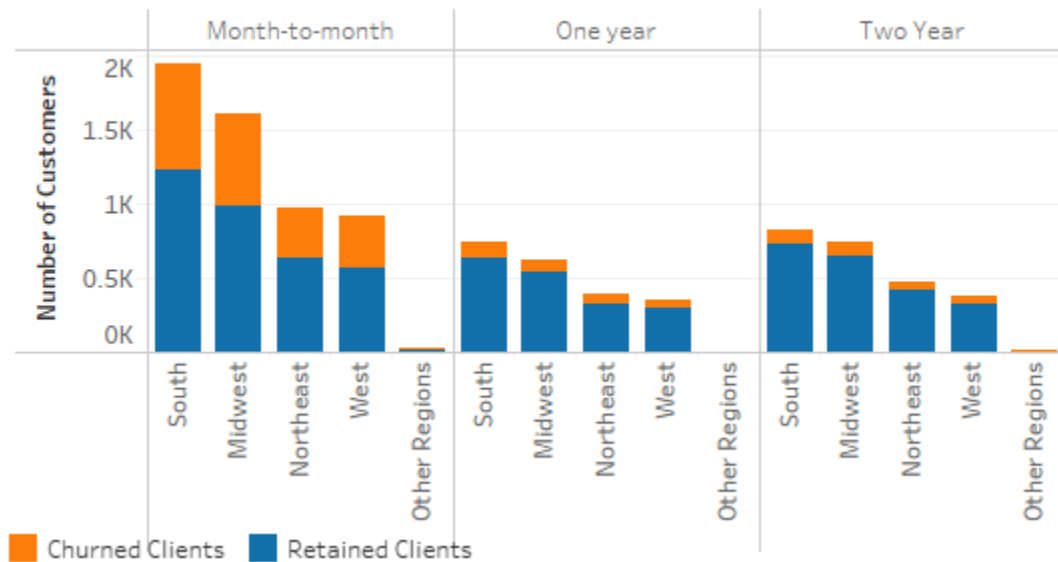


3. Understanding Key Visuals

Regional Analysis Dashboard

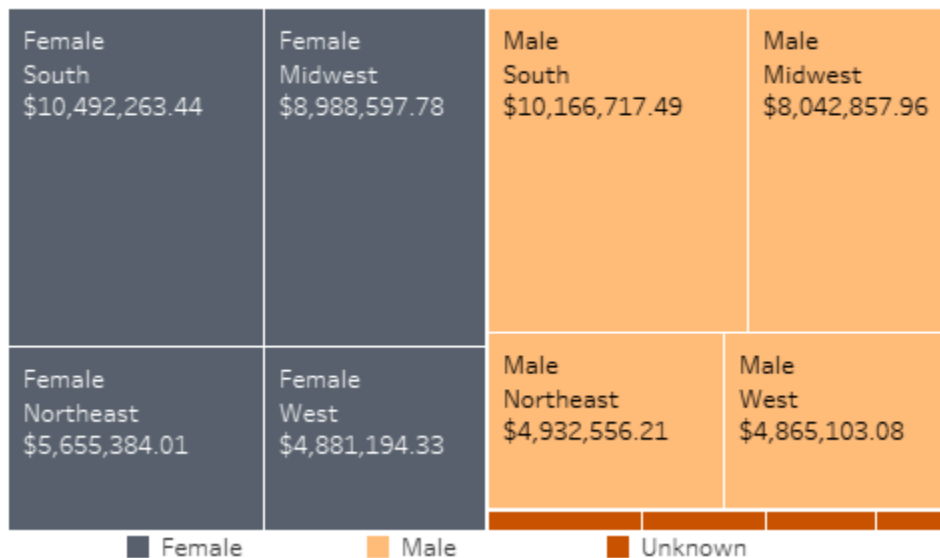
1. Stacked Bar Chart: Number of customers per region by contract type, colored by the customers who have been retained and the customers who have churned

Churn count by Contract Type

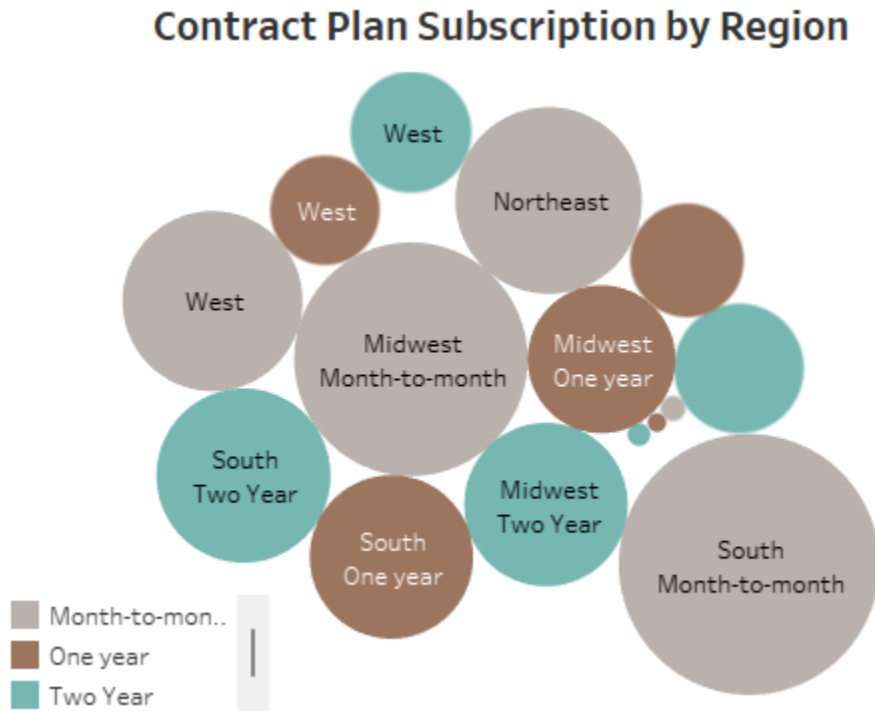


2. Side-by-Side Bar Chart: Compares churn and retention rates by region, with a tooltip for churn and retention rates by state.
3. Heatmap: Displays sales revenue by gender with regional insights with a tooltip for sales and monthly charges by state.

Regional Sales Revenue by Gender

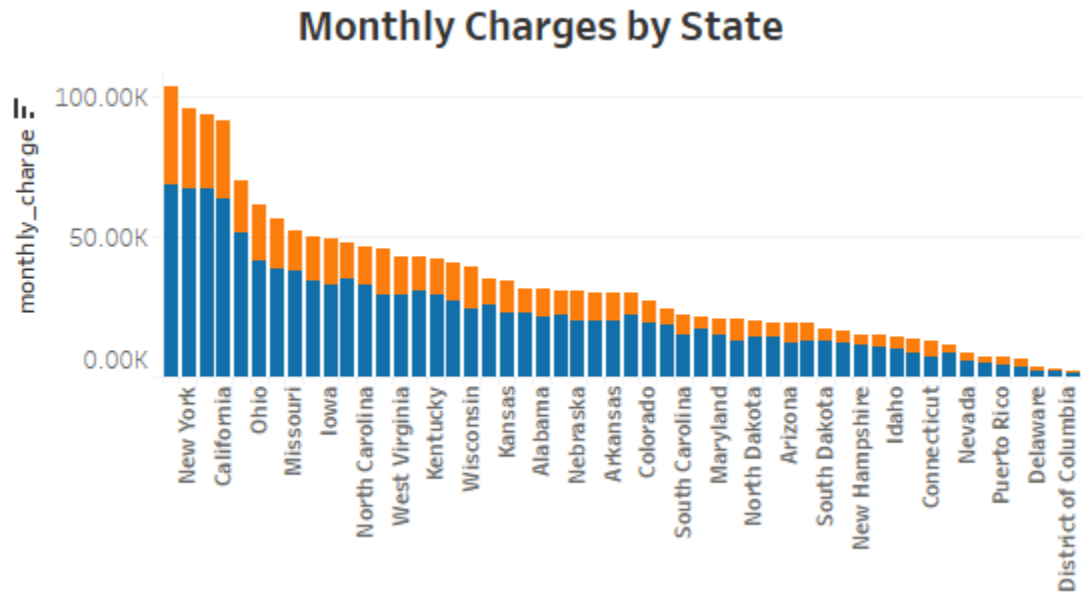


4. Bubble Chart: Represents contract subscription by region. With a tooltip that shows the customer count, sales amount, and average monthly charge

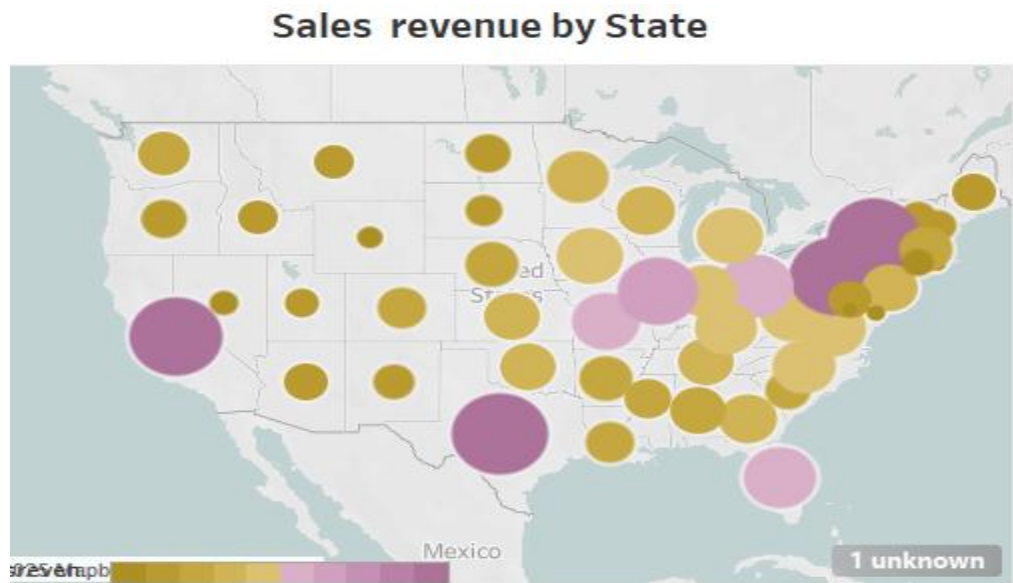


Sales Analysis Dashboard

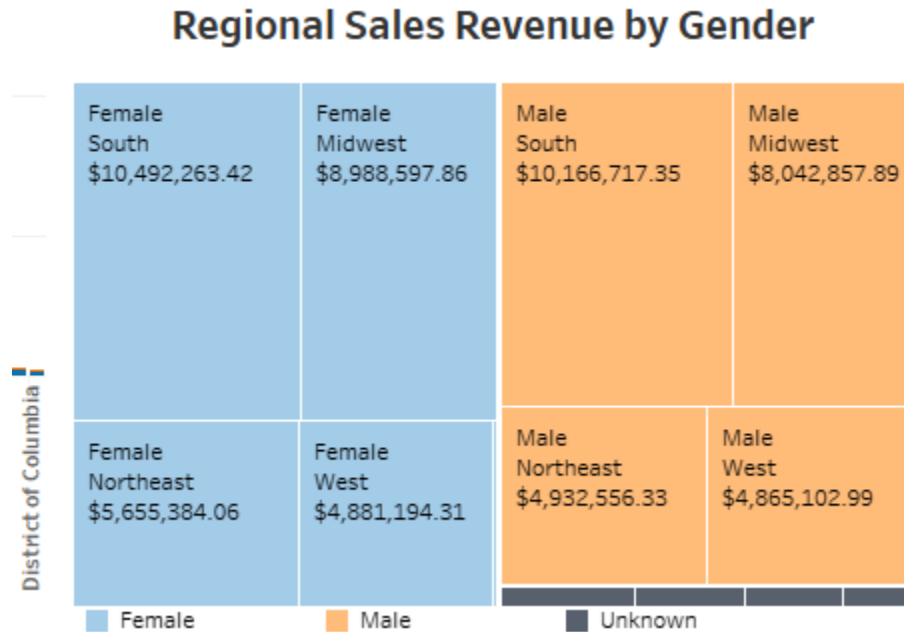
1. Stacked Bar Chart: Shows the Monthly charge by state, colored by the customers who have been retained and the customers who have churned with a tooltip that shows the monthly charges and average monthly charge.



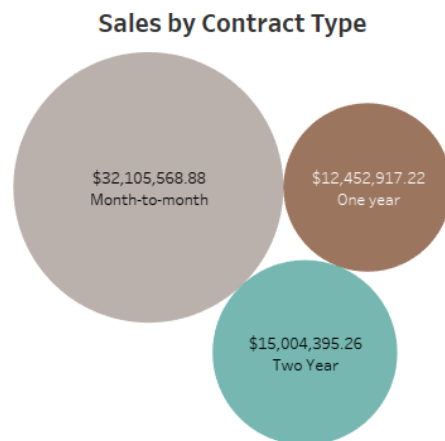
2. Map: Displays the total sales revenues that have been collected from each state. The total sales revenue is computed by multiplying the monthly charges by the tenure of the customer.



3. Heatmap: Displays sales revenue by gender with regional insights with a tooltip for sales and monthly charges by state.

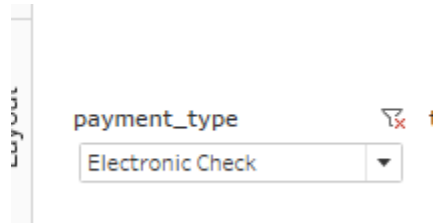


4. Bubble Chart: Represents sales by contract subscription with a tooltip showing the number of customers, the average monthly charge for the contract subscription, and the total monthly charges recovered for the customer

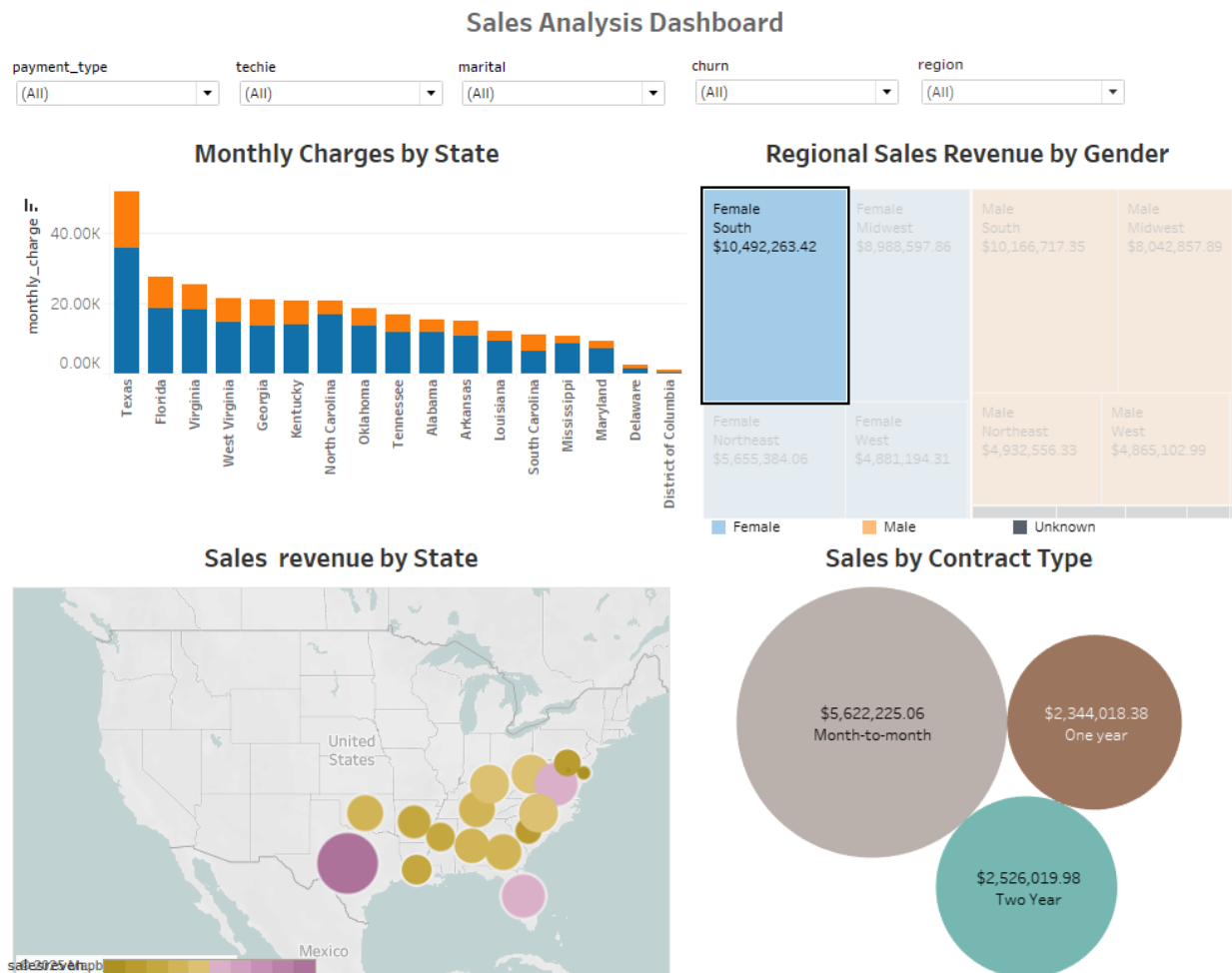


4. Resetting Filters

Click the Reset Filters funnel or manually deselect filters to return to the default view.

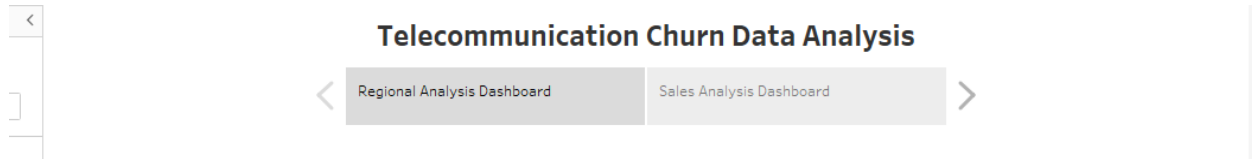


To reset an interactive visual, click on the same data point that was used to filter the data or click on a blank space on the visual.



5. Navigating Between Dashboards

Use the tabs at the top of the dashboard to switch between the views (Sales Analysis dashboard and Region Analysis dashboard).



A4: SQL CODE

SQL code for the creation of the dashboard

SQL code for the creation of the dashboard

```
SELECT
    cu.customer_id,
    cu.marital,
    CASE WHEN cu.churn = 'Yes' THEN 'Churned Clients' ELSE 'Retained Clients' END
    AS churn ,
    CASE WHEN cu.gender ='Prefer not to answer' THEN 'Unknown' ELSE cu.gender END
    AS gender,
    ROUND((cu.tenure* cu.monthly_charge),2) AS SalesRevenue,
    ROUND(cu.monthly_charge,2) AS monthly_charge ,
    CASE WHEN cu.techie ='Yes' THEN 'Has Techie Support' ELSE 'No TechieSupport '
    END AS techie,
    co.duration AS "Contract type",
    lo.city,
    lo.state,
```

```
lo.county,  
  
pa.payment_type,  
  
CASE WHEN sta.state ISNULL AND lo.state='PR' THEN 'Puerto Rico' ELSE sta.state  
  
END AS State_name,  
  
COALESCE( sta.region , 'Other Regions') AS region  
  
FROM customer cu  
  
LEFT JOIN contract co ON cu.contract_id=co.contract_id  
  
LEFT JOIN location lo ON cu.location_id=lo.location_id  
  
LEFT JOIN payment pa ON cu.payment_id=pa.payment_id  
  
LEFT JOIN states sta ON lo.state=sta.state_code
```

B: PANOPTO PRESENTATION

<https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=a11714d9-56d6-400d-9ada-b2e3000679d1>

B1: TECHNICAL ENVIRONMENT

For this analysis, the technical environment consisted of two business intelligence

1. pgAdmin
2. Tableau

pgAdmin is a powerful open-source graphical interface used to manage and interact with PostgreSQL databases, providing tools for writing queries, managing tables, and monitoring performance. In this project, pgAdmin was used to connect to and manage a database called Churn, which contains structured churn data used for churn analysis. The platform allowed for

efficient querying, data exploration, and schema management, making it easier to prepare and organize the data for further analysis.

The dashboard was developed using Tableau Desktop, a powerful business intelligence tool widely used for interactive data visualization. Tableau's intuitive drag-and-drop interface enabled the seamless design of charts, graphs, and key performance indicators (KPIs), which were then integrated into a user-friendly dashboard tailored to business needs.

The primary data source was a PostgreSQL Churn database, which contained structured customer data relevant to churn analysis. To optimize performance and enhance responsiveness, a Tableau Extract (.hyper file) was created from the original database. This extract allowed Tableau to process and render large volumes of data more efficiently by reducing dependency on real-time database queries.

Using a data extract also facilitated: Faster dashboard load times, Offline analysis, enabling work on the dashboard without needing a constant connection to the PostgreSQL server.

The final dashboard story is provided in .twbx (Tableau Packaged Workbook) format, which includes the dashboard, data extract, and all associated assets for seamless sharing and deployment.

B2: DEMONSTRATE DASHBOARD FUNCTIONALITY

The dashboard is designed to provide a clear, interactive overview of customer behavior, sales performance, and regional trends within the telecommunications data. Its functionality supports data-driven decision-making through effective visualization and filtering capabilities.

1. Interactive Filters

Both the Sales Analysis and Regional Analysis dashboards include filters that allow users to refine and customize their views based on: Payment_type, Techie, Marital, Churn, and Region

These filters dynamically update all visuals, enabling targeted exploration of the data.

2. Interactive Visuals

All charts and maps on the dashboards are interactive, allowing users to hover over visuals to reveal tooltips that display additional context by selecting portions of visuals to highlight specific data and update related charts within the same dashboard. This interactivity enhances user engagement and promotes deeper analysis without navigating away from the dashboard.

3. Sales Analysis Dashboard Functionality

This dashboard provides insights into:

- **Stacked Bar Chart:** Shows the Monthly charge by state, colored by the customers who have been retained and the customers who have churned, with a tooltip that shows the monthly charges and average monthly charge.
- **Heatmap:** Displays sales revenue by gender with regional insights, with a tooltip for sales and monthly charges by state.
- **Map:** Displays the total sales revenues that have been collected from each state. The total sales revenue is computed by multiplying the monthly charges by the tenure of the customer.
- **Bubble Chart:** Represents sales by contract subscription with a tooltip showing the number of customers, the average monthly charge for the contract subscription, and the total monthly charges recovered for the customer.

Each visual responds to filters and selections to present a tailored view of the performance.

4. Regional Analysis Dashboard Functionality

This dashboard enables:

- Stacked Bar Chart: Number of customers who have per region by contract type, colored by the customers who have been retained and the customers who have churned
- Side-by-Side Bar Chart: Compares churn and retention rates by region, with a tooltip for churn and retention rates by state.
- Heatmap: Displays sales revenue by gender with regional insights, with a tooltip for sales and monthly charges by state.
- Bubble Chart: Represents contract subscription by region. With a tooltip that shows the customer count, sales amount, and average monthly charge

Interactive elements and filters ensure that users can isolate regional performance metrics quickly and clearly.

5. Story Functionality

The Tableau Story titled "Telecommunication churn Data Analysis" brings together the Sales and Regional dashboards:

- Page 1: Regional Analysis
- Page 2: Sales Analysis

This sequential layout provides a structured narrative that guides the user through the analysis journey of the Churn dataset.

B3: SQL SCRIPTS

SQL code for the creation of the dashboard



reportquery.sql

SELECT

```
    cu.customer_id,
    cu.marital,
    CASE WHEN cu.churn = 'Yes' THEN 'Churned Clients' ELSE 'Retained Clients' END
    AS churn ,
    CASE WHEN cu.gender ='Prefer not to answer' THEN 'Unknown' ELSE cu.gender END
    AS gender,
    ROUND((cu.tenure* cu.monthly_charge),2) AS SalesRevenue,
    ROUND(cu.monthly_charge,2) AS monthly_charge ,
    CASE WHEN cu.techie ='Yes' THEN 'Has Techie Support' ELSE 'No TechieSupport '
    END AS techie,
    co.duration AS "Contract type",
    lo.city,
    lo.state,
    lo.county,
    pa.payment_type,
    CASE WHEN sta.state ISNULL AND lo.state='PR' THEN 'Puerto Rico' ELSE sta.state
    END AS State_name,
    COALESCE( sta.region , 'Other Regions') AS region
```

FROM customer cu

LEFT JOIN contract co ON cu.contract_id=co.contract_id

LEFT JOIN location lo ON cu.location_id=lo.location_id

LEFT JOIN payment pa ON cu.payment_id=pa.payment_id

LEFT JOIN states sta ON lo.state=sta.state_code

The data in the database is a consolidation of

1. The telecommunication industry Churn dataset
2. External data from Kaggle USA States and regions (states.csv). The dataset has 51 rows and 4 features. The data has been loaded into a database table called states in the Churn dataset

In creating the dashboard SQL query, the fields selected in the script were chosen specifically to support the interactive visual elements of the dashboard. The script pulls data from the tables below

- customer (cu) – Core customer information. The fields
 - customer_id: Unique identifier for each customer. The field is used to create quick calculations for the number of customers.
 - marital: Marital status of the customer used for demographic segmentation.
 - churn: Indicates whether the customer has churned—critical for retention analysis.
 - gender: Gender of the customer used for demographic trends.
 - (cu.tenure * cu.monthly_charge) AS SalesRevenue: This is a calculated column that calculates estimated total revenue per customer by multiplying each customer's tenure by their monthly charge. The calculated column Sales revenue is important for customer lifetime value (CLV) analysis.
 - monthly_charge: used to calculate the average monthly charge and total monthly charge.
 - Techie: Indicates whether the customer has techie support or not

- contract (co) – Details about the customer's service contract. The contract table provides the duration of each customer's service agreement, and supports the Bubble Chart and Stacked Bar Chart visuals that analyze contract types across regions
- location (lo) – Geographic data of the customer. The location table contributes to the city, state, and county, enabling geographic insights.
- payment (pa) – Payment methods used by customers. The payment_type field from the payment table supports filtering and analysis based on customer payment preferences.
- states (sta) – Mapping of state codes to state names and regions.

The SQL script is designed to extract, consolidate, and prepare essential customer and regional data from the Churn database to support the development of an interactive, executive-level dashboard for the telecommunication industry. The script draws data primarily from the customer table, which contains core customer information, and enriches it with related data from four additional tables: contract, location, payment, and states. These tables are joined using LEFT JOIN clauses to ensure that all customer records are preserved even if related information is not available in the joined tables.

The data produced by this SQL query supports several interactive visualizations in the dashboard, such as:

- Side-by-Side Bar Charts comparing churn vs. retention by region and state.
- Stacked Bar Charts that categorize customers by contract type and retention status across regions.
- Heatmaps that display total and average sales revenue segmented by gender and state.
- Bubble Charts that visualize the distribution of customers and revenue by contract and region.

- Geographic Map that shows total revenue collected by state.

Interactive features such as tooltips, filters (by Payment Method, Tech, Marital Status, and Churn), and hover-over functionalities are powered by the structured data retrieved by this script. These features allow business users to explore metrics like total sales revenue, churn rates, average monthly charges, and customer distribution across regions with ease.

The SQL script plays a foundational role in shaping the dataset used to populate the dashboard. It ensures the data is well-structured, complete, and aligned with the dashboard's analytical goals, providing actionable insights for decision-makers in the telecommunications sector.

B4: DATA STREAMS

The data streams used to support the dashboard analysis were prepared through a structured and logical data integration and transformation process. This preparation involved sourcing, joining, enriching, and calculating relevant data fields from multiple tables within the Churn database and external data sources, ensuring the resulting dataset was clean, comprehensive, and ready for advanced visual analysis.

The primary data stream originated from the telecommunication churn dataset, which resides in the customer table. This table contains key customer-level data such as customer_id, marital status, churn indicator, gender, tenure, monthly_charge, and techie flag. To enhance this core dataset, additional data streams were brought in through relational joins with other tables—specifically, contract, location, payment, and states.

Each of these supporting tables added specific attributes:

The contract table contributed to the duration field, identifying the length of each customer's service agreement. This is essential for contract-based analysis and helps to evaluate the relationship between contract commitment and churn behavior.

The location table provided geographical context with city, state, and county fields, allowing for spatial analysis of customer patterns and sales revenue across different regions.

The payment table supplied the payment_type, enabling behavioral segmentation based on how customers choose to pay.

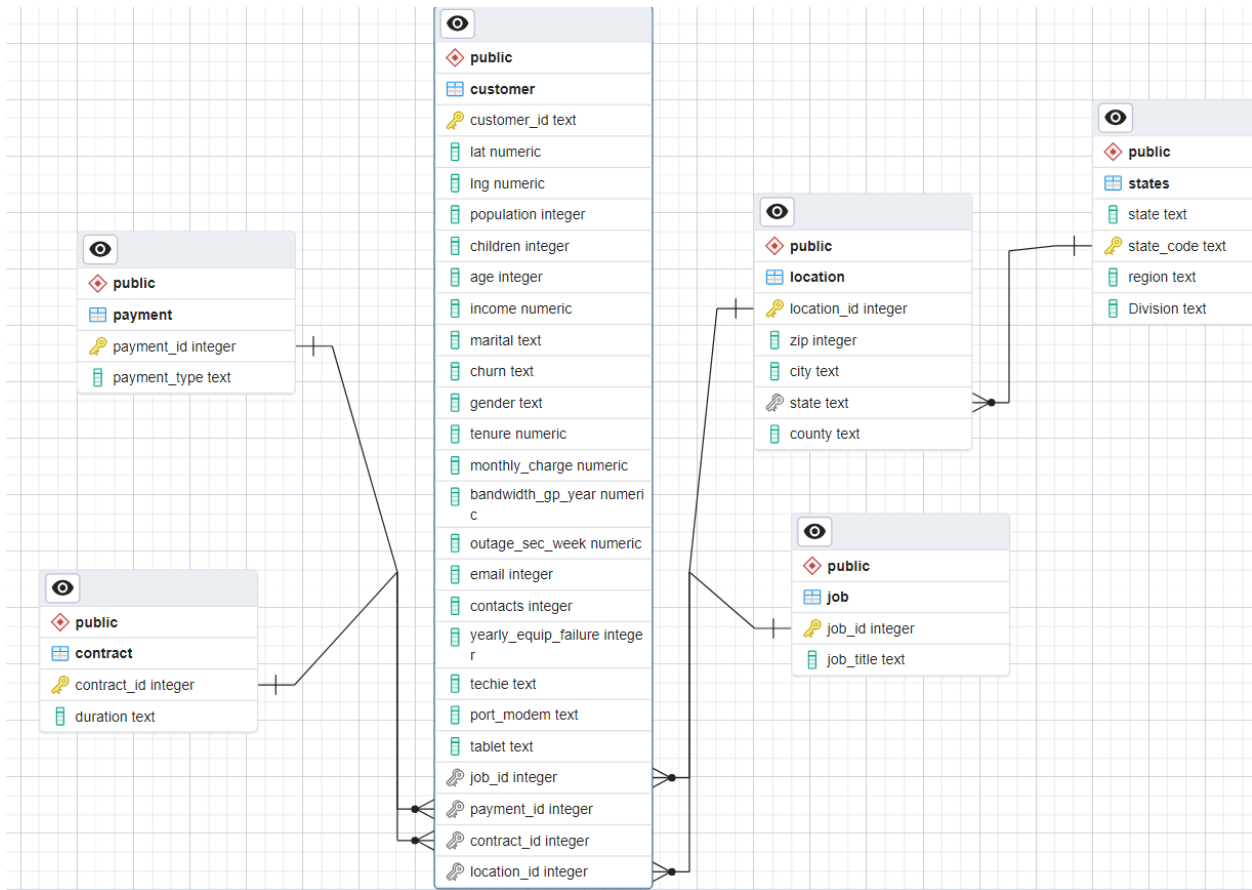
The states table, derived from external Kaggle data (states.csv), was integrated into the Churn database and linked via state_code. This enriched the geographic data with full state names and U.S. region classifications, which are vital for the regional breakdowns and filters in the dashboard.

To enable meaningful sales analysis, a calculated field, SalesRevenue, was created by multiplying each customer's tenure by their monthly_charge. This computed value reflects the total revenue generated by a customer over time and serves as a key performance metric across several dashboard visuals.

All data joins were performed using LEFT JOIN operations, ensuring that every customer was included in the final dataset, even if some related data was missing. This approach preserved the integrity of the customer base while maintaining the completeness of the analysis.

Once the data stream was fully prepared, it formed a flattened, analysis-ready dataset. This dataset was ideal for use in Tableau, where it powered various interactive dashboard visuals and filters allowing users to drill down and compare customer metrics from multiple perspectives.

The ERD diagram



B5: DATA POINTS

To ensure a cohesive and complete dataset for analysis, the data was aligned through a series of relational left joins that integrated key attributes from multiple sources. The primary dataset—the customer table—was enhanced by linking it with the contract, location, payment, and states tables using shared identifiers such as contract_id, location_id, payment_id, and state_code. This alignment brought together customer demographics, contract terms, payment types, and geographic information into a single, unified data stream. External regional data from the states table (sourced from Kaggle) was matched with the location data using standardized state codes, enabling consistent regional analysis. Additionally, a calculated field called SalesRevenue was derived by combining tenure and monthly charge values from within the customer table. This

integration ensured all data points were logically connected, complete, and ready for dashboard visualization across demographic, financial, behavioral, and geographic dimensions.

B6: DATABASE CREATION

The Churn database was created to support a comprehensive telecommunications churn analysis. It was designed using a relational schema that connects customer data with contract, payment, location, and regional details. The process of building the database included creating tables, establishing relationships through primary and foreign keys, and importing both internal and external datasets. Below is a step-by-step demonstration of how the database was created on my local PostgreSQL machine. The approach used ensured a clean, normalized, and referentially sound database capable of supporting complex analytical dashboards in Tableau. Each table was thoughtfully constructed to align with business goals and data relationships, making the Churn database both efficient and scalable.

1. Database Initialization

A new PostgreSQL database was created and named Churn

```
CREATE DATABASE churn
WITH
OWNER = postgres
ENCODING = 'UTF8'
LC_COLLATE = 'English_United States.1252'
LC_CTYPE = 'English_United States.1252'
LOCALE_PROVIDER = 'libc'
TABLESPACE = pg_default
CONNECTION LIMIT = -1
IS_TEMPLATE = False;
```

2. Table Creation

Tables were created based on the data model, each with appropriate data types and constraints.

The data types were chosen based on the nature of the data and to optimize performance and storage efficiency.

- Customer Table

```
CREATE TABLE IF NOT EXISTS public.customer
(
    customer_id text COLLATE pg_catalog."default" NOT NULL,
    lat numeric,
    lng numeric,
    population integer,
    children integer,
    age integer,
    income numeric,
    marital text COLLATE pg_catalog."default",
    churn text COLLATE pg_catalog."default",
    gender text COLLATE pg_catalog."default",
    tenure numeric,
    monthly_charge numeric,
    bandwidth_gp_year numeric,
    outage_sec_week numeric,
    email integer,
    contacts integer,
    yearly_equip_failure integer,
    techie text COLLATE pg_catalog."default",
    port_modem text COLLATE pg_catalog."default",
    tablet text COLLATE pg_catalog."default",
    job_id integer,
    payment_id integer,
    contract_id integer,
    location_id integer,
    CONSTRAINT customer_pkey PRIMARY KEY (customer_id),
    CONSTRAINT customer_contract_id_fkey FOREIGN KEY (contract_id)
        REFERENCES public.contract (contract_id) MATCH SIMPLE
        ON UPDATE NO ACTION
        ON DELETE NO ACTION,
    CONSTRAINT customer_job_id_fkey FOREIGN KEY (job_id)
        REFERENCES public.job (job_id) MATCH SIMPLE
        ON UPDATE NO ACTION
        ON DELETE NO ACTION,
    CONSTRAINT customer_location_id_fkey FOREIGN KEY (location_id)
        REFERENCES public.location (location_id) MATCH SIMPLE
        ON UPDATE NO ACTION
        ON DELETE NO ACTION,
    CONSTRAINT customer_payment_id_fkey FOREIGN KEY (payment_id)
        REFERENCES public.payment (payment_id) MATCH SIMPLE
        ON UPDATE NO ACTION
        ON DELETE NO ACTION
)

TABLESPACE pg_default;

ALTER TABLE IF EXISTS public.customer
    OWNER to postgres;
```

- Contract Table

```
CREATE TABLE IF NOT EXISTS public.contract
(
    contract_id integer NOT NULL,
    duration text COLLATE pg_catalog."default",
    CONSTRAINT contract_pkey PRIMARY KEY (contract_id)
)

TABLESPACE pg_default;

ALTER TABLE IF EXISTS public.contract
    OWNER to postgres;
```

- Location Table


```
CREATE TABLE IF NOT EXISTS public.location
(
    location_id integer NOT NULL,
    zip integer,
    city text COLLATE pg_catalog."default",
    state text COLLATE pg_catalog."default",
    county text COLLATE pg_catalog."default",
    CONSTRAINT location_pkey PRIMARY KEY (location_id),
    CONSTRAINT "State_fkey" FOREIGN KEY (state)
        REFERENCES public.states (state_code) MATCH SIMPLE
        ON UPDATE NO ACTION
        ON DELETE NO ACTION
        NOT VALID
)

TABLESPACE pg_default;

ALTER TABLE IF EXISTS public.location
    OWNER to postgres;
```

- Payment Table

```
CREATE TABLE IF NOT EXISTS public.payment
(
    payment_id integer NOT NULL,
    payment_type text COLLATE pg_catalog."default",
    CONSTRAINT payment_pkey PRIMARY KEY (payment_id)
)

TABLESPACE pg_default;

ALTER TABLE IF EXISTS public.payment
    OWNER to postgres;
```

- Job table

```
CREATE TABLE IF NOT EXISTS public.job
(
    job_id integer NOT NULL,
    job_title text COLLATE pg_catalog."default",
    CONSTRAINT job_pkey PRIMARY KEY (job_id)
)

TABLESPACE pg_default;

ALTER TABLE IF EXISTS public.job
    OWNER to postgres;
```

- states Table (from external Kaggle dataset)

```
CREATE TABLE IF NOT EXISTS public.states
(
    state text COLLATE pg_catalog."default",
    state_code text COLLATE pg_catalog."default" NOT NULL,
    region text COLLATE pg_catalog."default",
    "Division" text COLLATE pg_catalog."default",
    CONSTRAINT "Statecode_pkey" PRIMARY KEY (state_code)
)

TABLESPACE pg_default;

ALTER TABLE IF EXISTS public.states
    OWNER to postgres;
```

3. Data Import

Datasets were loaded into the respective tables using GUI-based tools in pgAdmin.

D211: AdvancedDataAcquisition_TableauDataDashboard
BeatriceM_WGU_MSDA35

Import/Export data - table 'contract'

General Options Columns

Import/Export ☒ Import ☐ Export

Filename C:\Users\beatr\OneDrive\Desktop\WGU\D211\D211DATA\contract.csv

Format csv

Encoding UTF8

Import/Export data - table 'job'

General Options Columns

Import/Export ☒ Import ☐ Export

Filename C:\Users\beatr\OneDrive\Desktop\WGU\D211\D211DATA\job.csv

Format csv

Encoding UTF8

Import/Export data - table 'payment'

General Options Columns

Import/Export ☒ Import ☐ Export

Filename C:\Users\beatr\OneDrive\Desktop\WGU\D211\D211DATA\payment.csv

Format csv

Encoding UTF8

Import/Export data - table 'states'

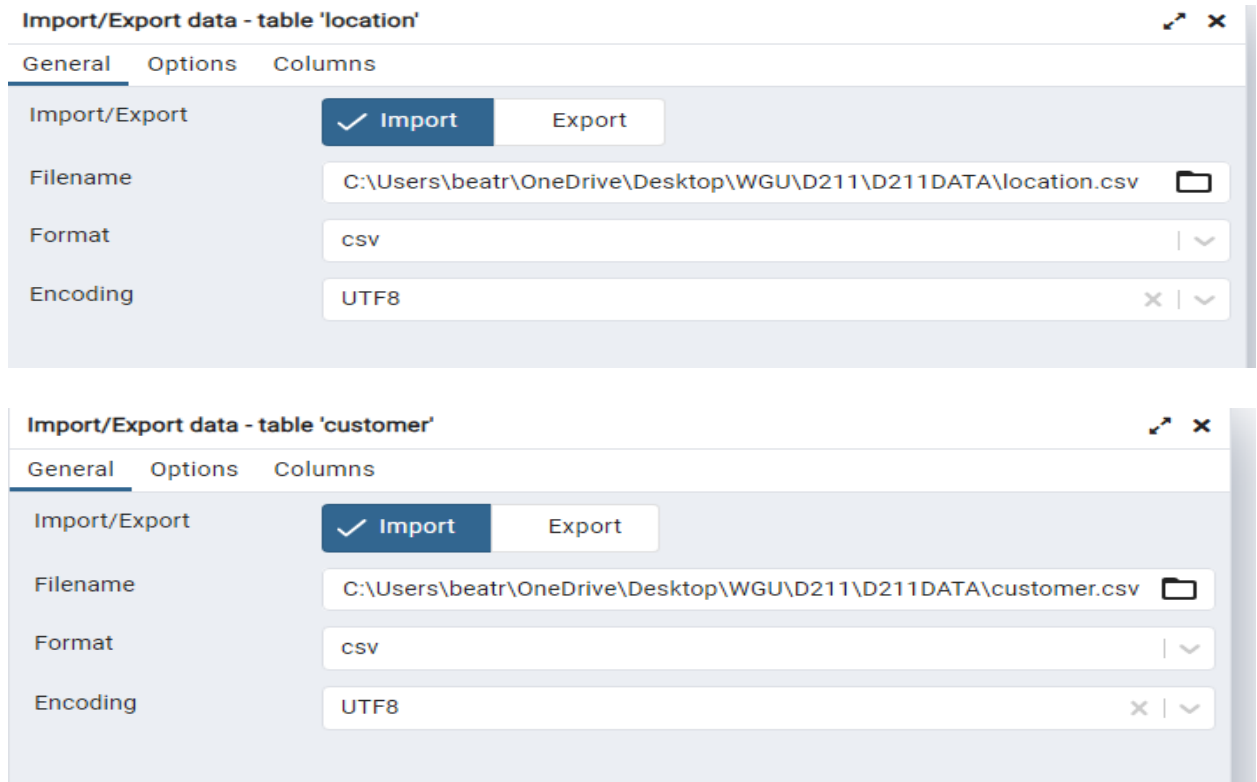
General Options Columns

Import/Export ☒ Import ☐ Export

Filename C:\Users\beatr\OneDrive\Desktop\WGU\D211\Externaldata\states.csv

Format csv

Encoding UTF8



The image displays two screenshots of the Tableau 'Import/Export data' dialog box, one for the 'location' table and one for the 'customer' table. Both dialog boxes are in the 'General' tab. The 'Import/Export' section shows 'Import' selected with a checkmark, and 'Export' is also visible. The 'Filename' field shows the path 'C:\Users\beatr\OneDrive\Desktop\WGU\D211\D211DATA\location.csv' for the first dialog and 'C:\Users\beatr\OneDrive\Desktop\WGU\D211\D211DATA\customer.csv' for the second. The 'Format' is set to 'csv' and the 'Encoding' is set to 'UTF8' in both. The 'Columns' tab is also visible in the background of each dialog.

4. Referential Integrity

Foreign key constraints were used to link the tables and enforce referential integrity. Each relationship uses ON UPDATE NO ACTION ON DELETE NO ACTION, meaning changes to or deletions of referenced data are restricted unless dependent records are first handled.

B7: REFERENTIAL INTEGRITY

Referential integrity in the Churn database was enforced through the implementation of foreign key constraints between related tables. These constraints ensure that relationships between tables remain valid and that data remains consistent throughout the database. Each foreign key establishes a link from one table's column (typically in a child table) to a corresponding primary key in another table (the parent table), preventing orphaned or mismatched records.

Here's how referential integrity was enforced across the key tables:

1. Customer Table as the Central Entity

The customer table serves as the main table and contains foreign keys that reference three supporting dimension tables:

- contract_id foreign key constraint references the table contract field contract_id
- location_id foreign key constraint references the table location field location_id
- payment_id foreign key constraint references the table payment field payment_id

```
CREATE TABLE IF NOT EXISTS public.customer
(
    customer_id text COLLATE pg_catalog."default" NOT NULL,
    lat numeric,
    lng numeric,
    population integer,
    children integer,
    age integer,
    income numeric,
    marital text COLLATE pg_catalog."default",
    churn text COLLATE pg_catalog."default",
    gender text COLLATE pg_catalog."default",
    tenure numeric,
    monthly_charge numeric,
    bandwidth_gp_year numeric,
    outage_sec_week numeric,
    email integer,
    contacts integer,
    yearly_equip_failure integer,
    techie text COLLATE pg_catalog."default",
    port_modem text COLLATE pg_catalog."default",
    tablet text COLLATE pg_catalog."default",
    job_id integer,
    payment_id integer,
    contract_id integer,
    location_id integer,
    CONSTRAINT customer_pkey PRIMARY KEY (customer_id),
    CONSTRAINT customer_contract_id_fkey FOREIGN KEY (contract_id)
        REFERENCES public.contract (contract_id) MATCH SIMPLE
        ON UPDATE NO ACTION
        ON DELETE NO ACTION,
    CONSTRAINT customer_job_id_fkey FOREIGN KEY (job_id)
        REFERENCES public.job (job_id) MATCH SIMPLE
        ON UPDATE NO ACTION
        ON DELETE NO ACTION,
    CONSTRAINT customer_location_id_fkey FOREIGN KEY (location_id)
        REFERENCES public.location (location_id) MATCH SIMPLE
        ON UPDATE NO ACTION
        ON DELETE NO ACTION,
    CONSTRAINT customer_payment_id_fkey FOREIGN KEY (payment_id)
        REFERENCES public.payment (payment_id) MATCH SIMPLE
        ON UPDATE NO ACTION
        ON DELETE NO ACTION
)

TABLESPACE pg_default;

ALTER TABLE IF EXISTS public.customer
    OWNER to postgres;
```

2. To associate geographic data with regional classifications, the location table's state column was linked to the states table using a foreign key. This constraint ensured that only valid U.S. state codes—those listed in the external Kaggle-based states table—could appear in the location table.

The state foreign key constraint references the table states field state_code

```
CREATE TABLE IF NOT EXISTS public.location
(
    location_id integer NOT NULL,
    zip integer,
    city text COLLATE pg_catalog."default",
    state text COLLATE pg_catalog."default",
    county text COLLATE pg_catalog."default",
    CONSTRAINT location_pkey PRIMARY KEY (location_id),
    CONSTRAINT "State_fkey" FOREIGN KEY (state)
        REFERENCES public.states (state_code) MATCH SIMPLE
        ON UPDATE NO ACTION
        ON DELETE NO ACTION
        NOT VALID
)

TABLESPACE pg_default;

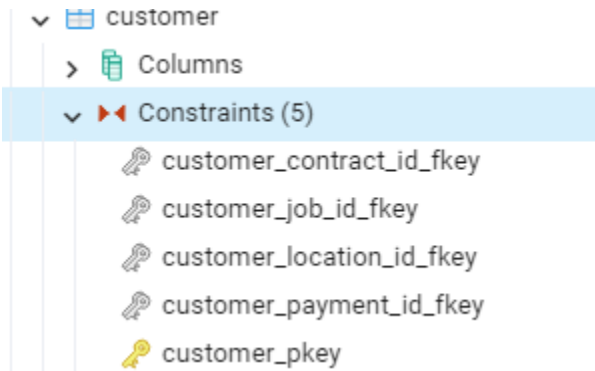
ALTER TABLE IF EXISTS public.location
    OWNER to postgres;
```

3. Primary Keys Defined for All Tables

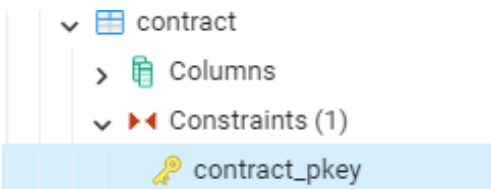
Each table was created with a clearly defined PRIMARY KEY. The Primary Key ensures uniqueness and provides reliable targets for foreign key references.

The primary Keys in the various tables are as below:

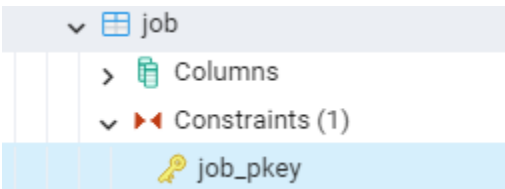
- PRIMARY KEY (customer_id) in the customer table



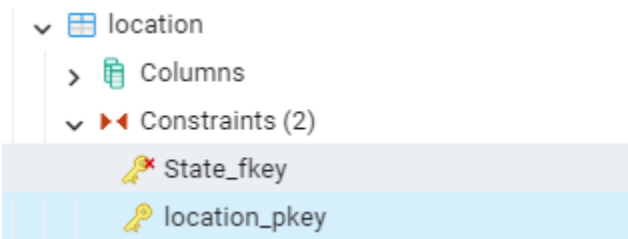
- PRIMARY KEY (contract_id) in the contract table



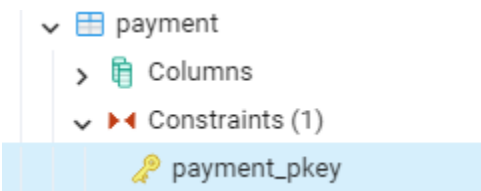
- PRIMARY KEY (job_id) on the job table



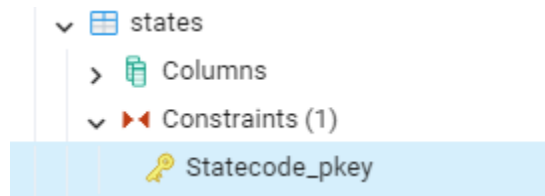
- PRIMARY KEY (location_id) in the location table



- PRIMARY KEY (payment_id) in the payment table



- PRIMARY KEY (state_code) in the states table



4. Prevention of Invalid Data Entry

With referential integrity enforced, the database automatically rejected any attempts to insert, update, or delete data that would violate foreign key relationships.

5. Optional ON DELETE/ON UPDATE Clauses

All foreign key constraints in the database were configured with ON DELETE NO ACTION and ON UPDATE NO ACTION. This means that if there is an attempt to delete or update a record in a parent table (such as contract, location, payment, or states) that is referenced by a child table (like customer), the database will prevent the action unless the referencing records are first modified or removed. This strict enforcement helps preserve data consistency by ensuring that no dependent records are left pointing to non-existent or changed entries.

C: WRITTEN REPORT

In the creation of the Regional and Sales Analysis Dashboard, the SQL script was developed to prepare the churn dataset for advanced analysis. Using PostgreSQL's capabilities such as CASE statements, calculated fields, conditional logic, rounding, joins, and null handling, the data was thoroughly cleaned, standardized, and enriched. This process transformed raw data into an analysis-ready format, providing a strong foundation for generating meaningful insights into customer churn, sales performance, and regional trends through interactive business intelligence dashboards

1. Data Exploration

The initial stage of the project involved exploring the customer churn data from a PostgreSQL-based dataset under the Churn database. The primary objective was to assess the structure, completeness, and consistency of the data and determine how it could be used to support meaningful business analysis.

To begin with, the total number of unique customers was calculated using the following SQL query: `SELECT COUNT(DISTINCT customer_id) FROM customer;`

This ensured that there were no duplicate customer records in the dataset. The uniqueness of customer IDs is critical for accurate aggregation and analysis of customer-level metrics.

The exploration process also included examining the range of attributes available in the core customer table, such as marital status, churn status, gender, tenure, and monthly charges, as well as inspecting the relationships with other tables such as contract, location, payment, payment, and a reference table named states which was imported from an external CSV file (Kaggle).

2. Use of Advanced SQL Operations

To prepare the dataset for analysis, several advanced SQL techniques were employed:

- Data Transformation Using CASE Statements

Several fields were recoded using CASE expressions to convert categorical values into more user-friendly and analysis-ready labels. For instance:

- Churn status was labeled as 'Churned Clients' and 'Retained Clients'.
- Gender values such as 'Prefer not to answer' were recoded to 'Unknown'.
- The tech-savvy indicator (techie) was converted into 'Has Techie Support' and 'No TechieSupport'.

- Calculated Fields

A calculated column called SalesRevenue was created by multiplying tenure by monthly_charge to estimate the total revenue generated per customer:

- Data Cleaning and Standardization
 - State codes such as 'PR' were manually aligned with full state names using a CASE statement:
 - Missing region values were handled using the COALESCE function:

- Joins for Data Integration

Data from multiple related tables was integrated using LEFT JOIN operations to preserve all customer records even if corresponding foreign key values were missing:

3. Analysis of the Data

The refined dataset enabled deeper analysis across several key business dimensions:

- Churn Analysis

By labeling churned vs. retained customers, I was able to analyze churn rates across segments such as contract type, region, and payment method.

- Revenue Insights

The SalesRevenue field enabled analysis of revenue contribution by contract type, state, gender, techie support, payment methods, and by region.

- Demographic and Behavioral Segmentation

The filters used fields like marital status, gender, techie, and payment methods, allowing segmentation of the customer base for better targeting and personalization strategies.

- Geographic Analysis

Using the state and region data, regional trends could be visualized to support location-based marketing and service delivery improvements.

C1: DASHBOARD ALIGNMENT

For this analysis, I chose the Churn dataset from the telecommunications industry, which provides critical customer and regional data needed to analyze customer retention and sales performance. The purpose of the dashboard is to deliver clear, actionable insights that support executive decision-making by highlighting key metrics such as churn rates, sales revenue, and customer behavior across regions. To meet stakeholder needs, the dashboard is organized into two complementary parts: the Regional Analysis Dashboard and the Sales Analysis Dashboard from my D210 performance assessment. The Regional Analysis Dashboard enables stakeholders to monitor customer retention and sales trends by region through various visualizations, including bar charts, heatmaps, and bubble charts. These visuals provide a logical and comprehensive view of where churn is highest and which contract types are most prevalent, helping executives identify areas requiring targeted interventions. The Sales Analysis Dashboard focuses on sales-related metrics, showing monthly charges, revenue by state, and contract subscription details. This supports stakeholders' need to understand the financial impact of customer churn and evaluate sales performance geographically and demographically. The dashboard's function is supported by carefully constructed SQL queries that extract and integrate relevant data from the Churn dataset provided by WGU and state data that was obtained externally from the kaggle.com website, ensuring that stakeholders have access to accurate and consolidated information. By combining intuitive visualizations with robust data integration, the dashboard fully aligns with stakeholder priorities for timely, data-driven insights that guide strategic decisions in reducing churn and maximizing revenue.

C2: BUSINESS INTELLIGENCE TOOL

For the purpose of this analysis, I chose Tableau Desktop because of its intuitive drag-and-drop interface and versatile integration capabilities, making it ideal for building effective and interactive visualizations. Tableau is a leading business intelligence tool widely recognized for its robust set of visualization options, flexibility in dashboard customization, and strong support community with regularly updated documentation. The dashboard was developed using data from a PostgreSQL Churn database, which contained structured customer information relevant to churn analysis. One of the key benefits of choosing Tableau is its ability to create data extracts (hyper files), which I utilized to optimize performance. By generating an extract from the original data source, I significantly improved dashboard load times, enabled offline access, and reduced dependency on real-time queries, making the analysis faster and more efficient. The final dashboard is delivered as a .twbx (Tableau Packaged Workbook), which includes the dashboard, extract, and all associated assets for seamless sharing and deployment.

C3: DATA CLEANING

To ensure the dataset was clean, consistent, and ready for accurate analysis and visualization, several key data preparation techniques were applied. These included resolving inconsistencies, transforming values into meaningful labels, handling nulls, and formatting numerical fields for clarity and precision:

1. Checking for duplicates.

I started by checking if there were any duplicates in the customer table by checking for unique values.

```
SELECT COUNT(DISTINCT customer_id) FROM customer;
```

Data Output Messages

≡	📄	▼	📋	▼	🗑️	📊	📥	📈
	count							
	bigint							
1	10000							

2. Resolving inconsistencies

- Replaced ambiguous gender values like 'Prefer not to answer' with 'Unknown'
- Standardized techie values for clarity from 'Yes'/'No' to 'Has Techie Support'/'No TechieSupport'

- Corrected unmatched state names for 'PR' mapped to 'Puerto Rico'

3. Transforming values into meaningful labels for end-user readability

- Converted churn values ('Yes'/'No') into descriptive categories ('Churned Clients'/'Retained Clients')
- Renamed fields like duration as "Contract type" to align with dashboard labels

4. Handling missing values

- Used CASE and COALESCE functions to replace missing state and region values with 'Puerto Rico' and 'Other Regions'
- Ensured completeness in key dimensions to avoid visualization gaps

5. Formatting numbers appropriately

- Rounded monthly_charge and computed SalesRevenue to two decimal places
- Ensured consistent numeric formatting for use in summaries, charts, and financial calculations

```
SELECT
cu.customer_id,
cu.marital,
CASE WHEN cu.churn = 'Yes' THEN 'Churned Clients' ELSE 'Retained Clients' END AS churn ,
CASE WHEN cu.gender = 'Prefer not to answer' THEN 'Unknown' ELSE cu.gender END AS gender,
ROUND((cu.tenure* cu.monthly_charge),2) AS SalesRevenue,
ROUND(cu.monthly_charge,2) AS monthly_charge ,
CASE WHEN cu.techie = 'Yes' THEN 'Has Techie Support' ELSE 'No TechieSupport' END AS techie ,
co.duration AS "Contract type",
lo.city,
lo.state,
lo.county,
pa.payment_type,
CASE WHEN sta.state ISNULL AND lo.state='PR' THEN 'Puerto Rico' ELSE sta.state END AS State_name,
COALESCE( sta.region , 'Other Regions') AS region
FROM customer cu
LEFT JOIN contract co ON cu.contract_id=co.contract_id
LEFT JOIN location lo ON cu.location_id=lo.location_id
LEFT JOIN payment pa ON cu.payment_id=pa.payment_id
LEFT JOIN states sta ON lo.state=sta.state_code
```

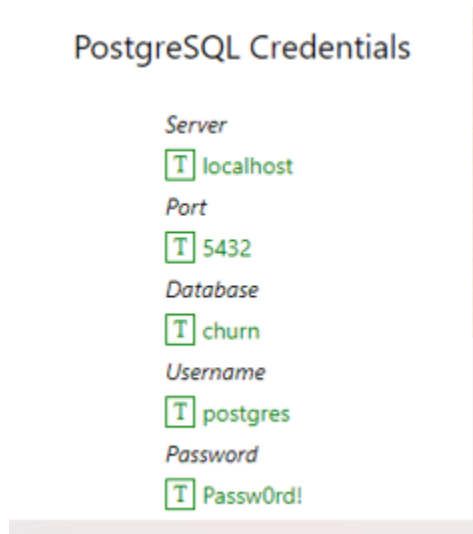
C4: DASHBOARD CREATION

Step 1: Prerequisites

- pgAdmin has been installed
- Churn dataset from the telecommunication industry, hosted in a PostgreSQL database via pgAdmin.
- Tableau desktop is loaded

Step 2: Loading External Data and Data Preparation

Log in the pgAdmin interface using the credentials below



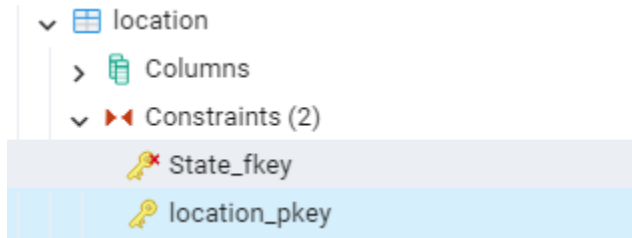
Create a database table called states in the Churn database as per the script below, and add a foreign key to the field state table location connecting it to the states table field state_code

```
CREATE TABLE IF NOT EXISTS public.states  
(  
    state text COLLATE pg_catalog."default",  
    state_code text COLLATE pg_catalog."default" NOT NULL,  
    region text COLLATE pg_catalog."default",  
    "Division" text COLLATE pg_catalog."default",  
    CONSTRAINT "Statecode_pkey" PRIMARY KEY (state_code)  
)
```

```
TABLESPACE pg_default;
```

```
ALTER TABLE IF EXISTS public.states
```

```
    OWNER to postgres;
```

Load data from external states data from Kaggle (USA states and regions) into the states table.

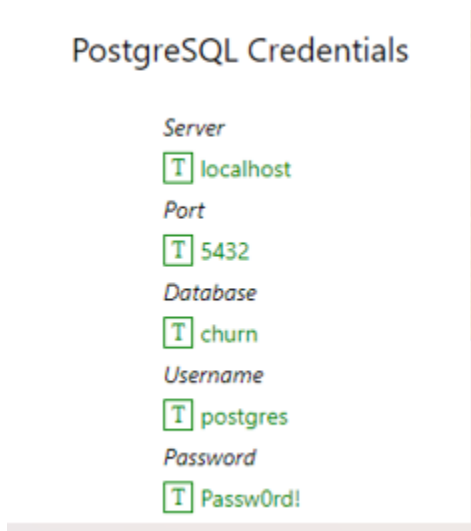
<https://www.kaggle.com/datasets/omer2040/usa-states-to-region/data>

see the attached file

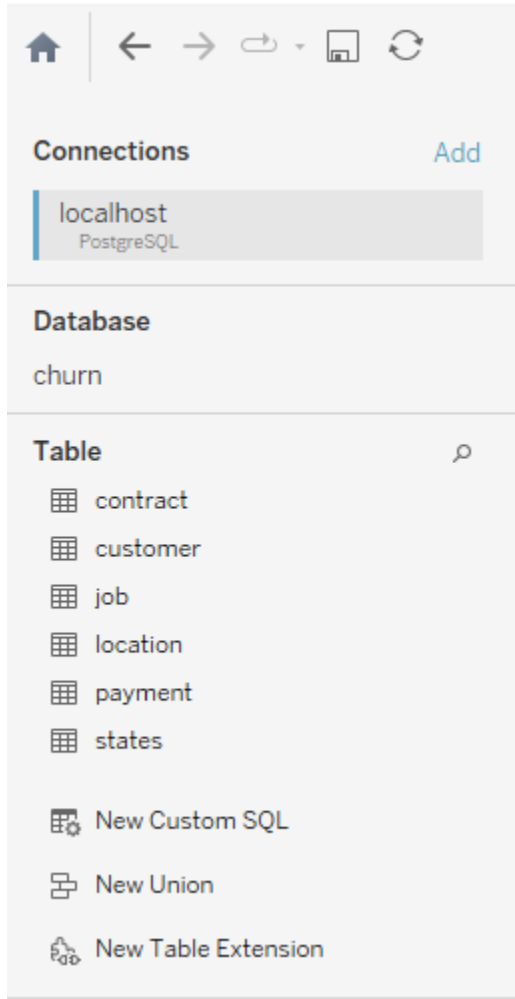


Step 3: Data Connection

Connect Tableau to the PostgreSQL Churn database. Use the credentials below



Click on New custom SQL



Construct the custom SQL query to retrieve the cleaned and consolidated dataset and name the custom query D211ReportingQuery. Create a Tableau Data Extract (.hyper file) to improve performance, enable offline access, and speed up dashboard loading.

```
SELECT
    cu.customer_id,
    cu.marital,
    CASE WHEN cu.churn = 'Yes' THEN 'Churned Clients' ELSE 'Retained Clients' END
    AS churn ,
```

```
CASE WHEN cu.gender ='Prefer not to answer' THEN 'Unknown' ELSE cu.gender END  
  
AS gender,  
  
ROUND((cu.tenure* cu.monthly_charge),2) AS SalesRevenue,  
  
ROUND(cu.monthly_charge,2) AS monthly_charge ,  
  
CASE WHEN cu.techie ='Yes' THEN 'Has Techie Support' ELSE 'No TechieSupport '  
  
END AS techie,  
  
co.duration AS "Contract type",  
  
lo.city,  
  
lo.state,  
  
lo.county,  
  
pa.payment_type,  
  
CASE WHEN sta.state ISNULL AND lo.state='PR' THEN 'Puerto Rico' ELSE sta.state  
  
END AS State_name,  
  
COALESCE( sta.region , 'Other Regions') AS region  
  
FROM customer cu  
  
LEFT JOIN contract co ON cu.contract_id=co.contract_id  
  
LEFT JOIN location lo ON cu.location_id=lo.location_id  
  
LEFT JOIN payment pa ON cu.payment_id=pa.payment_id  
  
LEFT JOIN states sta ON lo.state=sta.state_code
```

D211: AdvancedDataAcquisition_TableauDataDashboard BeatriceM_WGU_MSDA52

churn

Connection: Live Extract Edit Refresh Filters: 0 | Add
Extract contains subset of data. 5/18/2025 4:35:19 PM

D211ReportingQu...

Edit Custom SQL

```
SELECT
cu.customer_id,
cu.marital,
CASE WHEN cu.churn = 'Yes' THEN 'Churned Clients' ELSE 'Retained Clients' END AS churn ,
CASE WHEN cu.gender = 'Prefer not to answer' THEN 'Unknown' ELSE cu.gender END AS gender,
ROUND((cu.tenure* cu.monthly_charge),2) AS SalesRevenue,
ROUND(cu.monthly_charge,2) AS monthly_charge ,
CASE WHEN cu.techie = 'Yes' THEN 'Has Techie Support' ELSE 'No TechieSupport' END AS techie ,
co.duration AS "Contract type",
lo.city,
lo.state,
lo.county,
pa.payment_type,
CASE WHEN sta.state ISNULL AND lo.state='PR' THEN 'Puerto Rico' ELSE sta.state END AS State_name,
COALESCE( sta.region , 'Other Regions') AS region
FROM customer cu
LEFT JOIN contract co ON cu.contract_id=co.contract_id
LEFT JOIN location lo ON cu.location_id=lo.location_id
LEFT JOIN payment pa ON cu.payment_id=pa.payment_id
LEFT JOIN states sta ON lo.state=sta.state_code
```

Preview Results... Insert Parameter

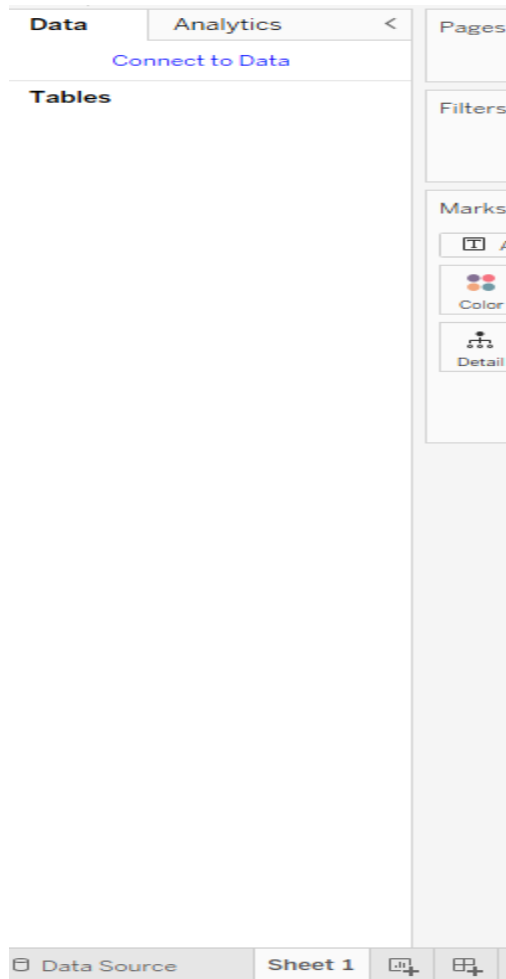
D211ReportingQuery 17 fields 10000 rows 100 rows

Name	Type	Field Name	Physical Table	Remote...	customer_id	marital	churn	gender	salesrevenue	monthly_charge	techie	Contract type	city	state
D211ReportingQuery	Custom SQL Query	customer_id	customer_id		K409198	Widowed	Retained Clients	Male	\$1171.92	\$172.46	No TechieSupport	One year	Cailhoun	IL
	Custom SQL Query	marital	marital		S120509	Married	Churned Clients	Female	\$280.65	\$242.63	Has Techie Support	Month-to-month	Neptune Beach	FL
	Custom SQL Query	churn	churn		K191035	Widowed	Retained Clients	Female	\$2,519.84	\$159.95	Has Techie Support	Two Year	Linesville	PA
	Custom SQL Query	gender	gender		D90850	Married	Retained Clients	Male	\$2,049.73	\$119.96	Has Techie Support	Two Year	Dawson	ND
	Custom SQL Query	salesrevenue	salesrevenue		K662701	Separated	Churned Clients	Male	\$250.56	\$149.95	No TechieSupport	Month-to-month	Sabana Seca	PR
	Custom SQL Query	monthly_charge	monthly_charge		W303516	Never Married	Retained Clients	Female	\$1,295.24	\$185.01	No TechieSupport	One year	London	KY
	Custom SQL Query	techie	techie		U335188	Widowed	Churned Clients	Male	\$2,648.92	\$200.12	Has Techie Support	Month-to-month	Burton	MI
	Custom SQL Query	Contract type	Contract type		V538685	Married	Churned Clients	Female	\$490.18	\$114.95	Has Techie Support	Month-to-month	Whiteface	TX
	Custom SQL Query				M716771	Separated	Retained Clients	Unknown	\$965.67	\$117.47	No TechieSupport	Month-to-month	New Hampton	NH
	Custom SQL Query				I676080	Married	Retained Clients	Female	\$556.03	\$162.48	No TechieSupport	Two Year	Ottertail	MN
	Custom SQL Query				J080369	Separated	Retained Clients	Female	\$3,370.96	\$174.96	No TechieSupport	Month-to-month	Mc Lemoresv...	TN
	Custom SQL Query				E243720	Married	Retained Clients	Female	\$1,577.90	\$149.96	No TechieSupport	Month-to-month	Dix	IL

Step 4: creating Worksheets and formatting

- Create calculated columns

Open a new worksheet

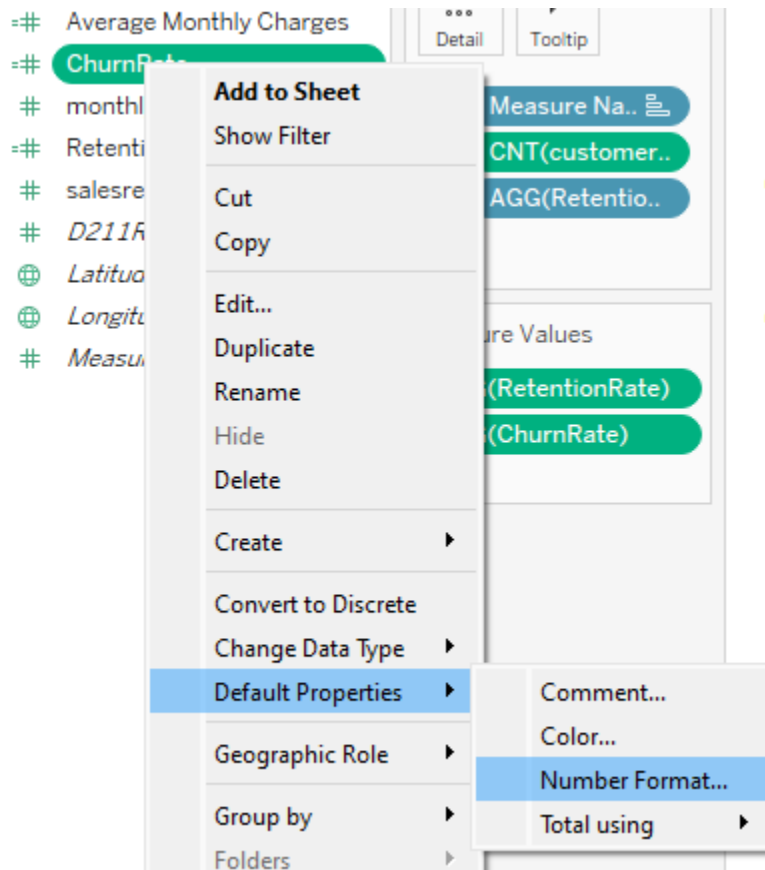


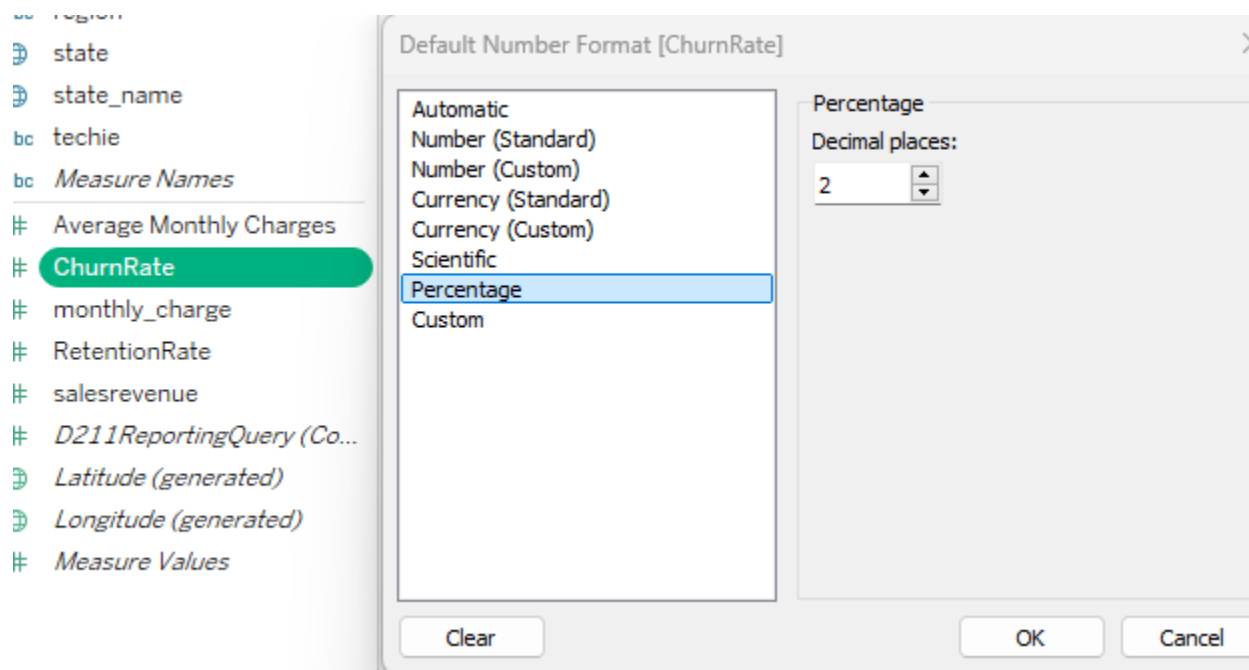
Go to Data Pane > Right-click > Create Calculated Field as below then click apply

1. Churn Rate

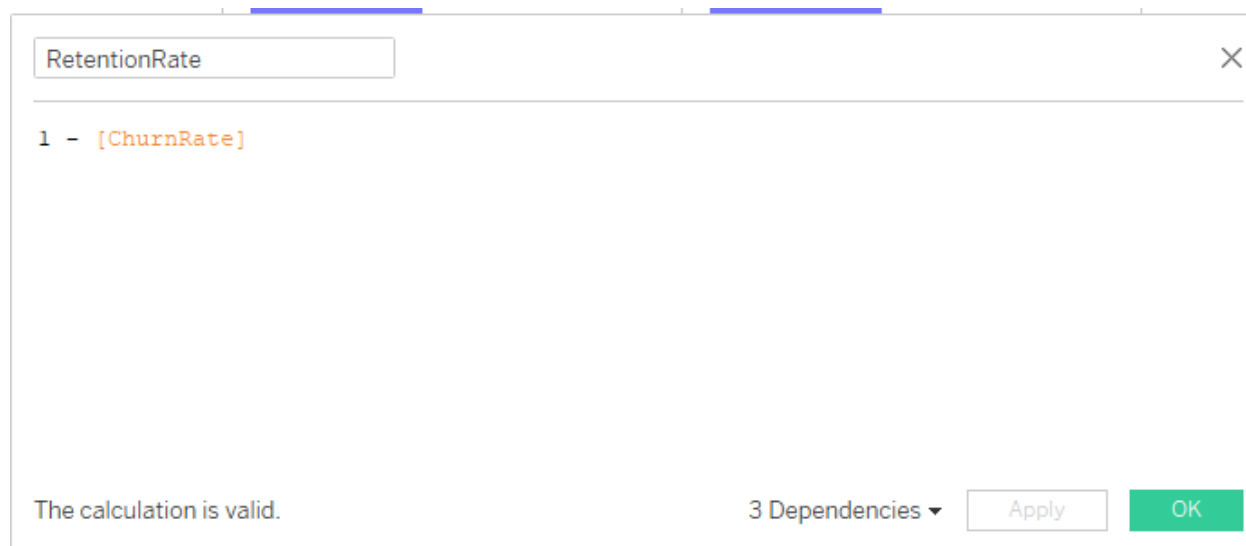


Right click on the measure and set the Default properties >Number properties to percentage

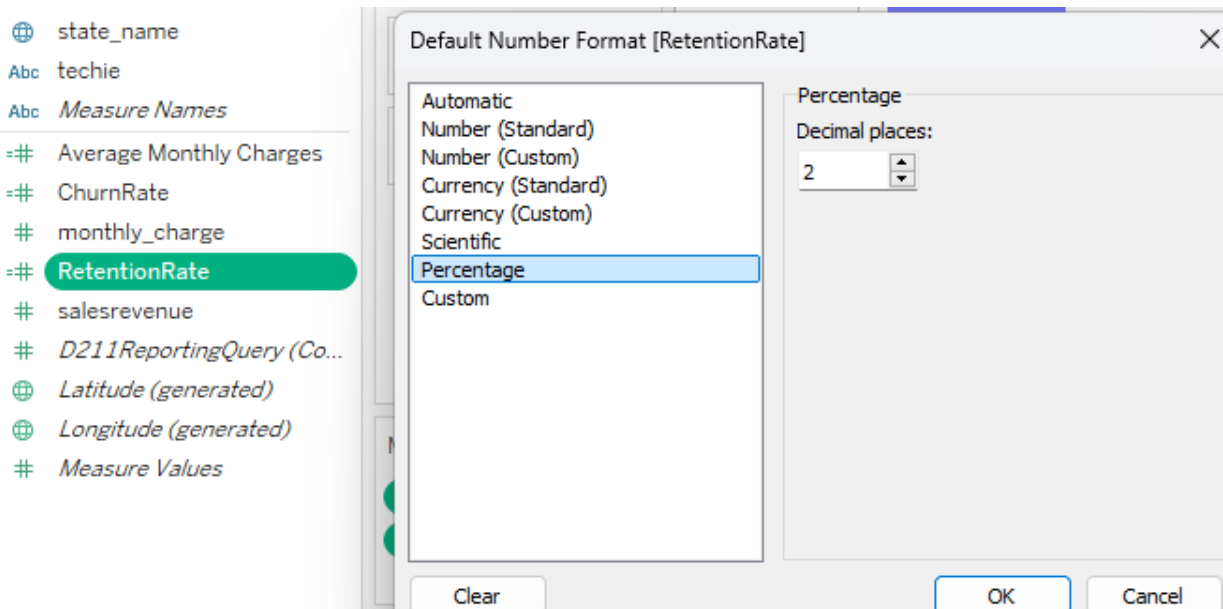
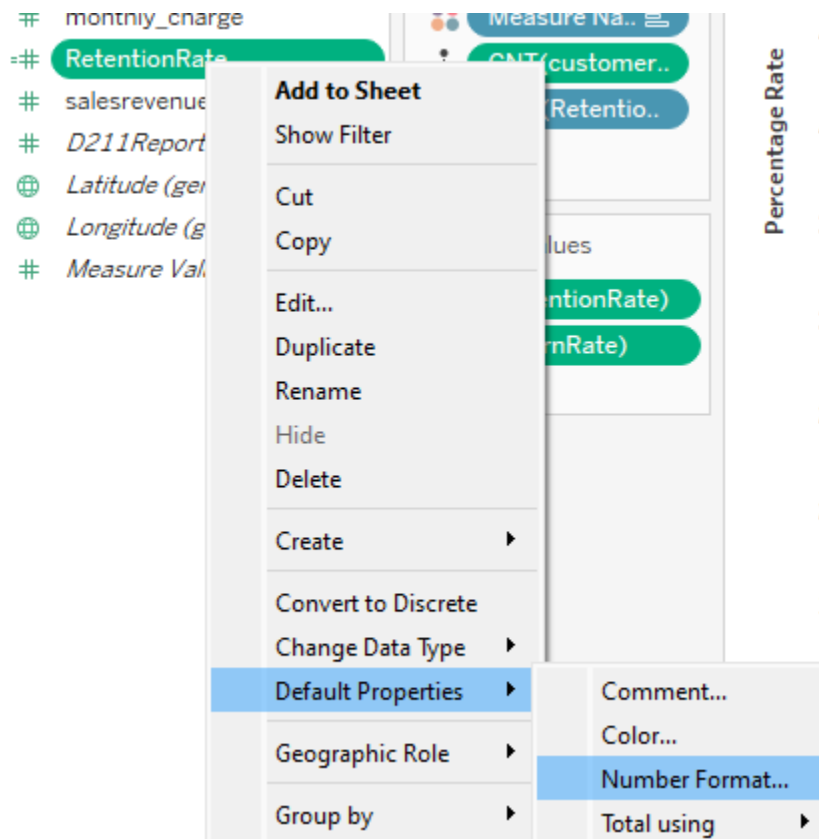




2. Retention Rate



Right click on the measure and set the Default properties >Number properties to percentage



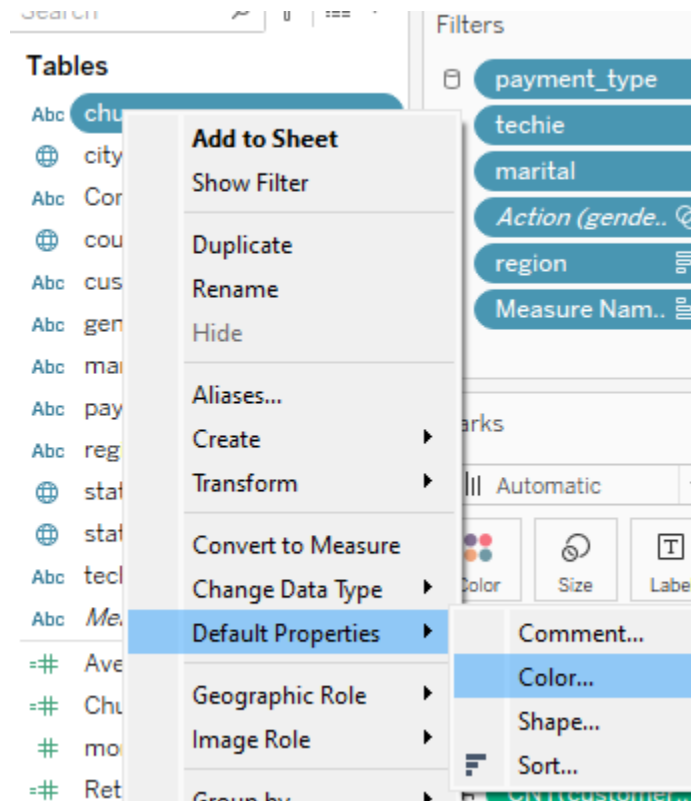
3. Average Monthly Charge

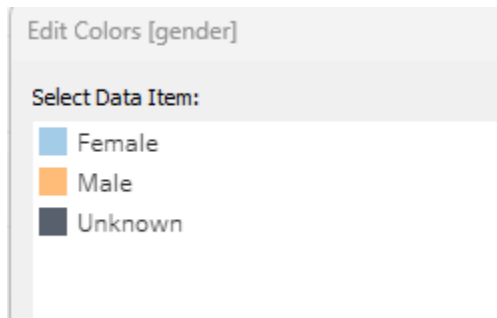
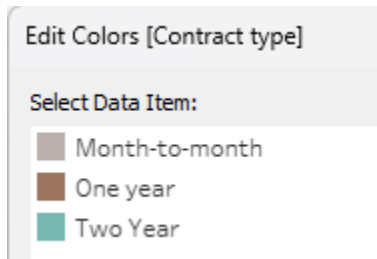
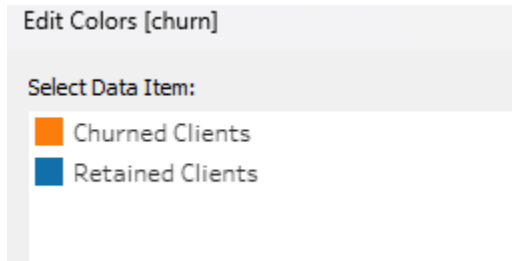


- **Set default color values**

Set default colors for Churn, Contract type and Gender

Right click on the field > default properties> color then set the color as below

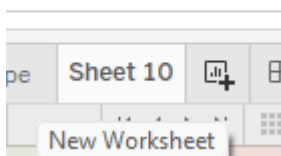




- **Create Worksheets**

Each worksheet is an individual chart or table used in the dashboard or as part of the tooltip.

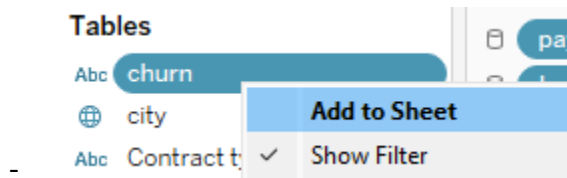
To create a new worksheet, click on the icon below



Filter formatting for all visuals and all Filters

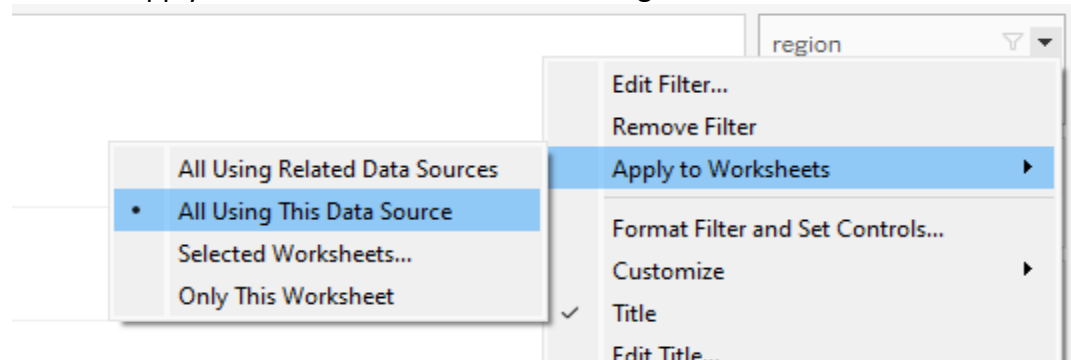
- Filters to be created for payment type, techie, marital, churn, and region on each of the worksheets that has filters

- Right-click on each dimension above and click Show Filter

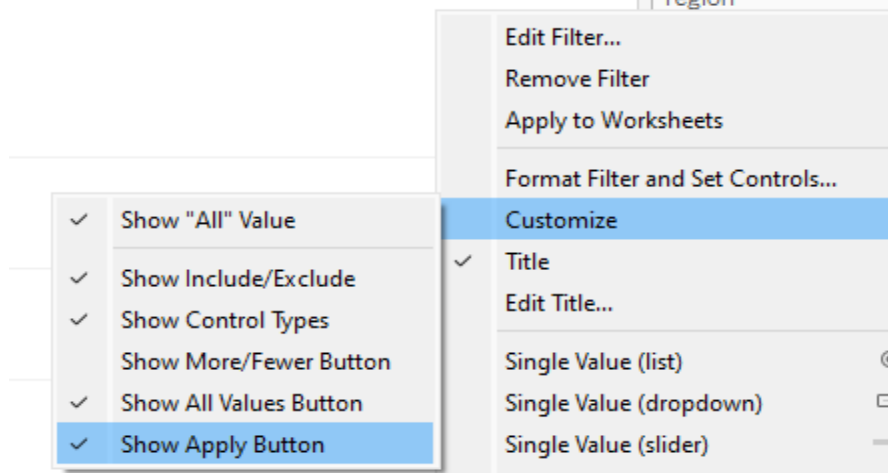


- Once the filter is added on the right-hand side, right-click on the arrow and format as below

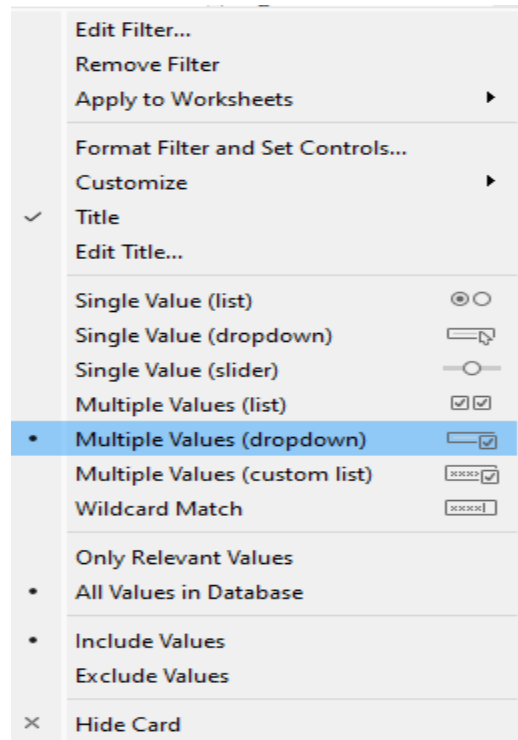
- o Click on Apply to worksheets and select All Using this data source



- o Select Customize and click on the Show Apply button



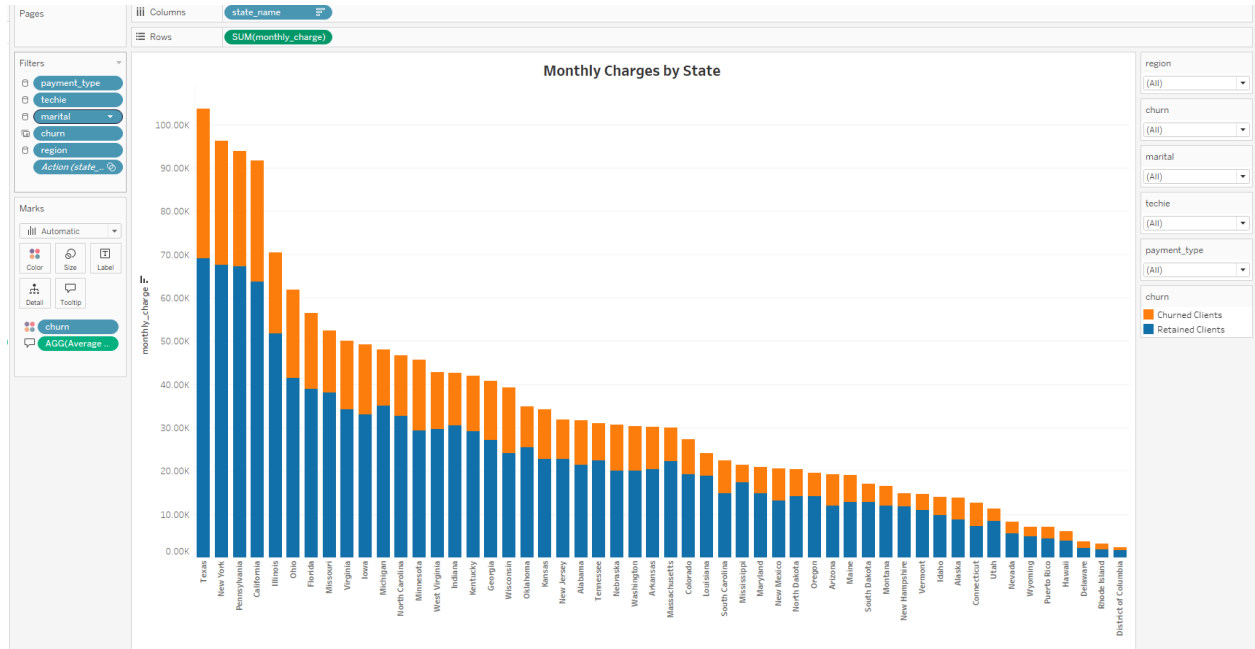
- o Select Multiple values drop-down



Worksheet 1: Stacked Bar – Monthly Charge by State (Churned vs. Retained)

1. Drag State to Columns.
2. Drag monthly_charge to Rows. This will default in SUM. If it does not, right-click on monthly_charge and set Measure to SUM.
3. Drag churn to Color. This will format in the default colors for churned and retained
4. Add Tooltip info: average monthly charge, monthly charge, format the state, and the Churned vs retained measure
5. Sort states by total monthly charge (descending).
6. Center the Title and Change View to entire View
7. Add the filters for payment type, techie, marital, churn and region

D211: AdvancedDataAcquisition_TableauDataDashboard BeatriceM_WGU_MSDA61



Worksheet 2: Table – Sales by State

1. Drag State to Rows.
2. Add SalesRevenue and monthly_charge to the Columns.
3. Set view to Fit Height
4. Hide the title

D211: AdvancedDataAcquisition_TableauDataDashboard BeatriceM_WGU_MSDA62

Pages

Columns

Measure Names

Rows

state_name

Filters

Tooltip (regio. ...)

Measure Nam. B.

Tooltip (gend. ...)

payment_type

techie

marital

churn

region

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Measure Values

Measure Values

SUM(salesrevenue)

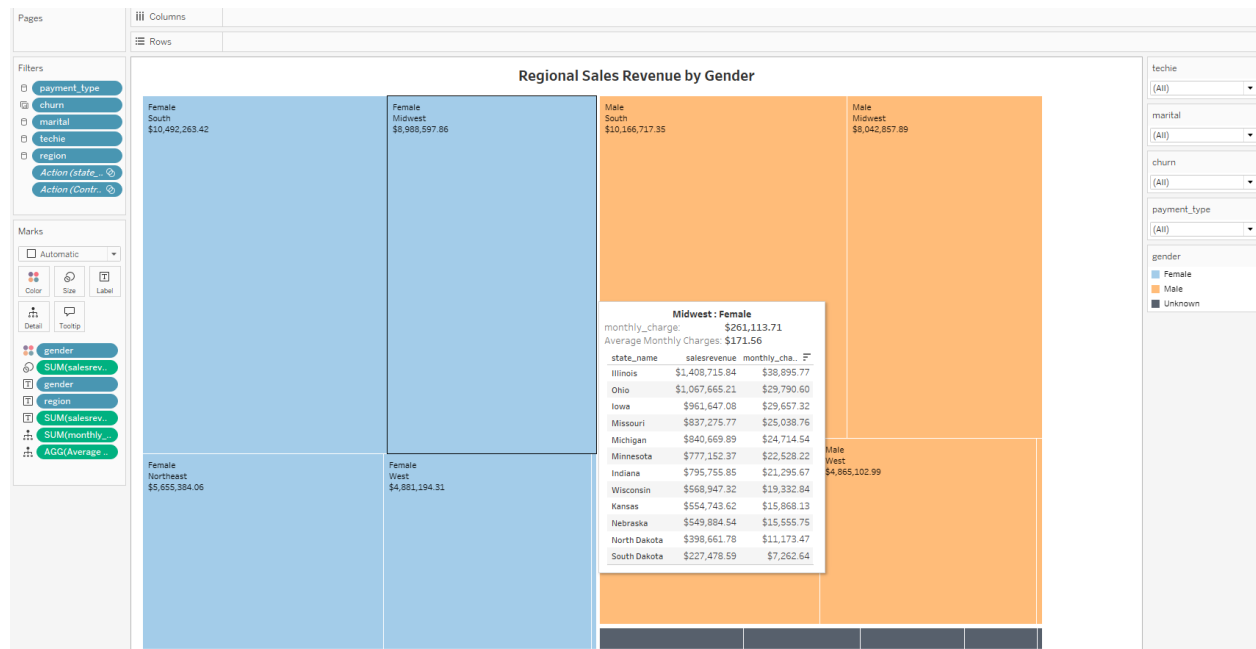
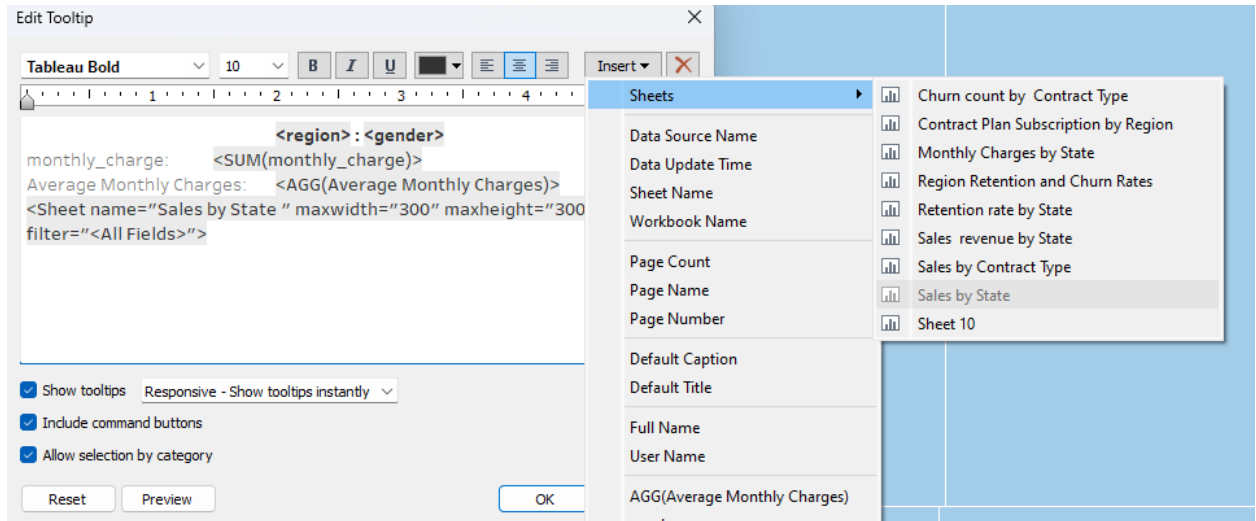
SUM(monthly_charg.

state_name	salesrevenue	monthly_cha.
Texas	\$3,496,988.77	\$103,712.84
New York	\$3,243,719.32	\$96,234.15
Pennsylvania	\$3,256,572.47	\$93,955.46
California	\$3,257,688.56	\$91,677.89
Illinois	\$2,451,368.57	\$70,474.28
Ohio	\$2,130,226.76	\$61,833.58
Florida	\$2,004,529.35	\$56,442.36
Missouri	\$1,804,921.89	\$52,407.75
Virginia	\$1,761,959.24	\$50,079.26
Iowa	\$1,659,292.90	\$49,213.74
Michigan	\$1,662,326.36	\$47,986.95
North Carolina	\$1,551,017.35	\$46,711.98
Minnesota	\$1,433,967.10	\$45,619.51
West Virginia	\$1,530,189.10	\$42,771.62
Indiana	\$1,477,445.03	\$42,632.31
Kentucky	\$1,465,138.06	\$41,929.35
Georgia	\$1,340,213.08	\$40,786.12
Wisconsin	\$1,267,159.83	\$39,285.17
Oklahoma	\$1,156,200.71	\$34,920.29
Kansas	\$1,150,071.09	\$34,198.00
New Jersey	\$1,121,642.08	\$31,790.87
Alabama	\$1,093,965.71	\$31,755.19
Tennessee	\$1,159,636.03	\$31,089.61
Nebraska	\$1,071,711.47	\$30,614.64
Washington	\$1,001,176.79	\$30,421.84
Arkansas	\$1,084,648.37	\$30,119.11
Massachusetts	\$1,060,421.69	\$29,930.75
Colorado	\$929,525.83	\$27,343.42
Louisiana	\$895,157.74	\$24,183.72
South Carolina	\$782,372.48	\$22,432.01
Mississippi	\$823,758.88	\$21,467.86
Maryland	\$720,377.65	\$20,934.20
New Mexico	\$646,247.51	\$20,534.86
North Dakota	\$748,135.62	\$20,404.05
Oregon	\$742,110.50	\$19,626.51
Arizona	\$662,061.64	\$19,261.18
Maine	\$678,464.45	\$18,989.13
South Dakota	\$515,662.55	\$16,942.84
Montana	\$592,600.76	\$16,478.52
New Hampshire	\$535,968.35	\$14,760.42
Vermont	\$517,039.72	\$14,696.12
Idaho	\$544,200.49	\$14,023.60
Alaska	\$414,944.73	\$13,790.96
Connecticut	\$389,121.84	\$12,616.94
Utah	\$437,382.90	\$11,278.34
Nevada	\$312,375.94	\$8,305.64
Wyoming	\$251,670.39	\$7,159.82
Puerto Rico	\$254,469.38	\$7,070.15
Hawaii	\$169,289.12	\$6,120.47
Delaware	\$132,519.36	\$3,675.84
Rhode Island	\$90,650.23	\$3,132.73
District of Columbia	\$82,175.62	\$2,385.23

Worksheet 3: Heatmap – Regional Sales Revenue by Gender

1. Drag the Gender to the labels on the Marks
2. Drag the Region to the labels on the Marks
3. Drag SalesRevenue to size. on the Marks
4. Click on the Show Me tab on the right-hand side and choose the heatmap visual
5. Drag the gender to the Color tab on the Marks
6. Drag monthly_charge to the Tooltip, add Worksheet two to the tooltip, and format

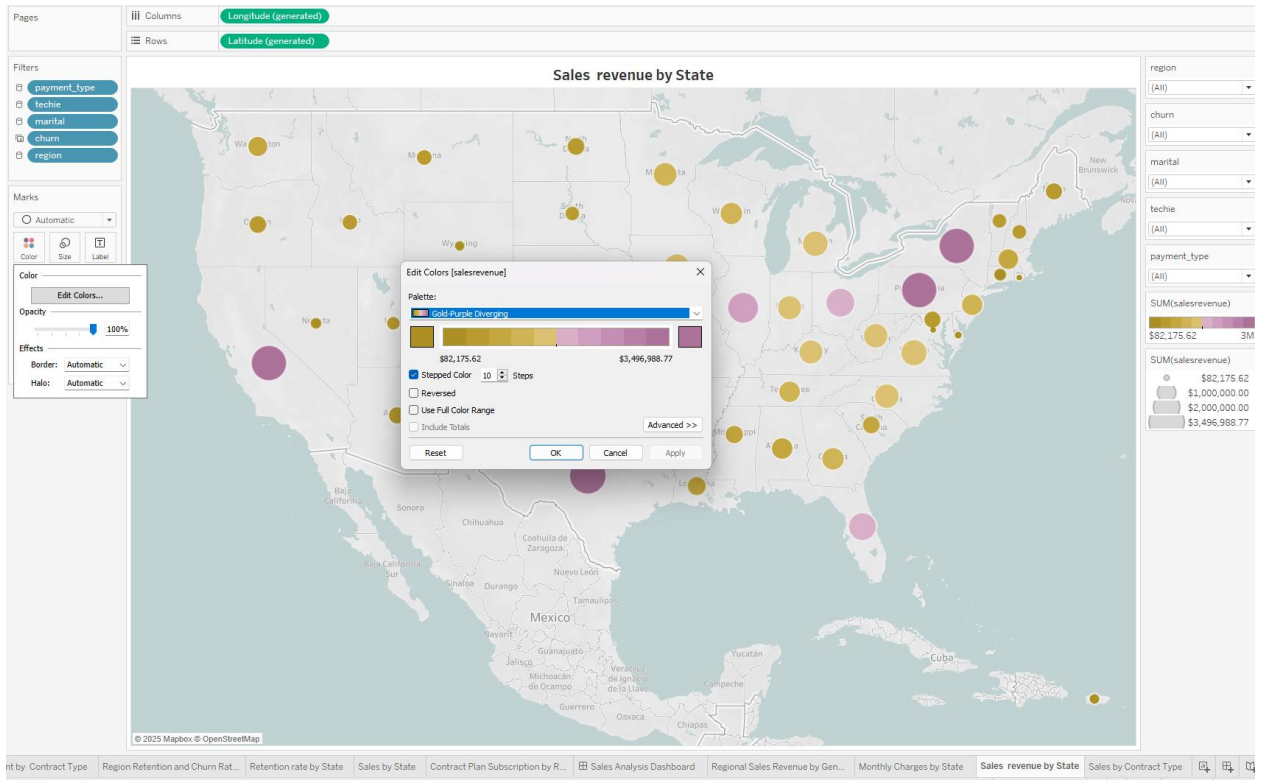
D211: AdvancedDataAcquisition_TableauDataDashboard BeatriceM_WGU_MSDA63



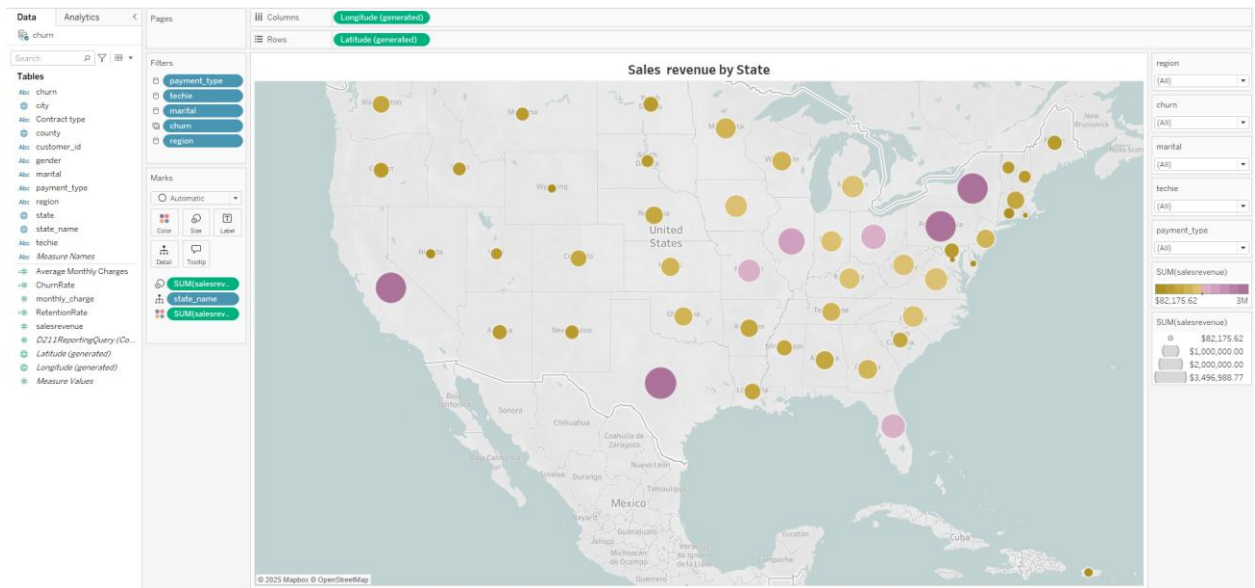
Worksheet 4: Map – Total Sales Revenue by State

1. Drag the Longitude on the columns field and the Latitude on the rows field
2. Drag the Sales revenue to the size tab on the Marks card
3. Drag the Sales revenue to the color tab on the Marks card and format the colors as below

D211: AdvancedDataAcquisition_TableauDataDashboard BeatriceM_WGU_MSDA64



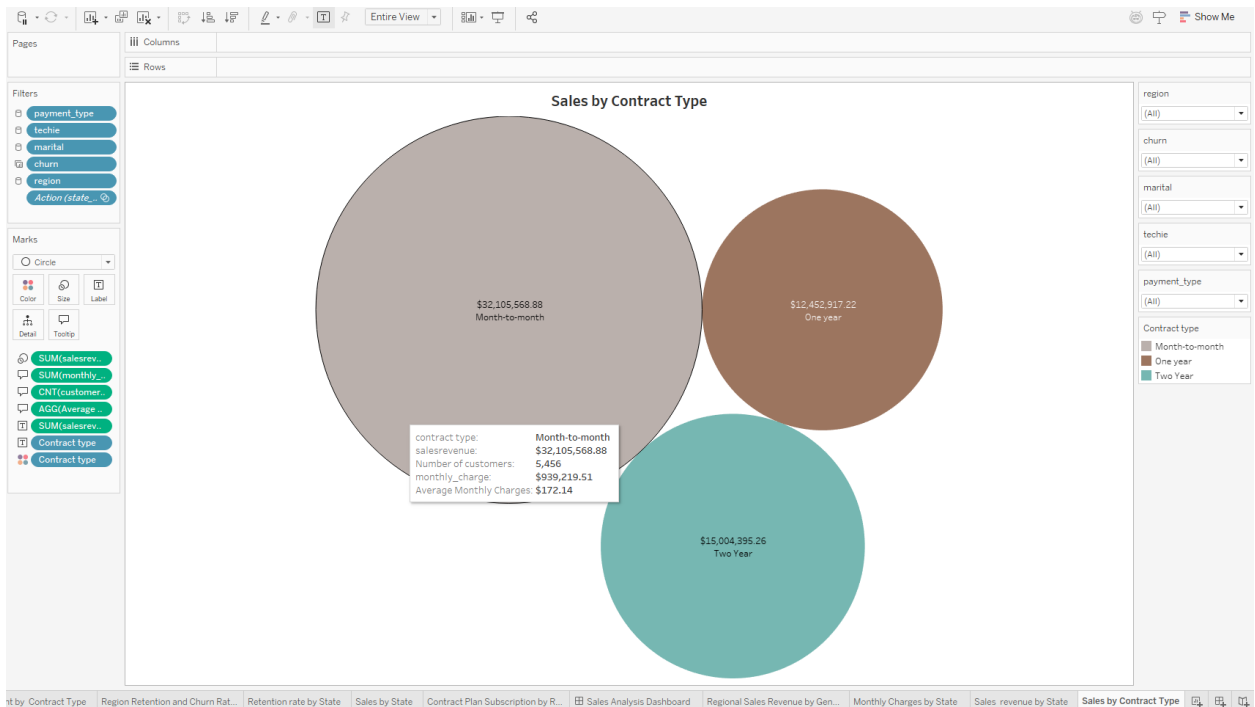
4. Add the filters as shown below



Worksheet 5: Bubble Chart – Sales by Contract Type

1. Drag the Contract to the labels on the Marks
2. Drag SalesRevenue to size On the Marks and ensure that the measure is set to SUM

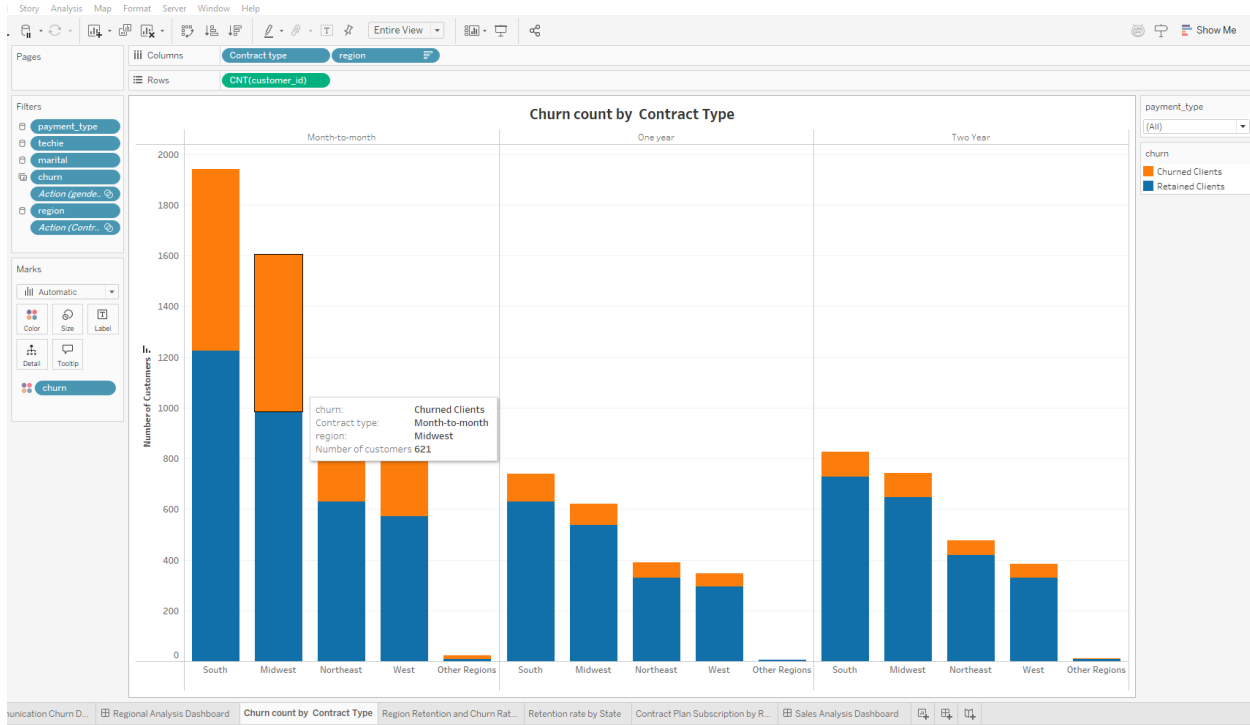
3. Click on the Show Me tab on the right-hand side and choose the bubble chart visual
4. Drag the Contract Type to the color.
5. Drag SalesRevenue to Size.
6. Drag average monthly_charge to Tooltip and customer_id (count) to Tooltip.
7. Add the filters as shown below
8. Format the title and change the view to Entire view



Worksheet 6: Stacked Bar – Churn count by contract type and Region

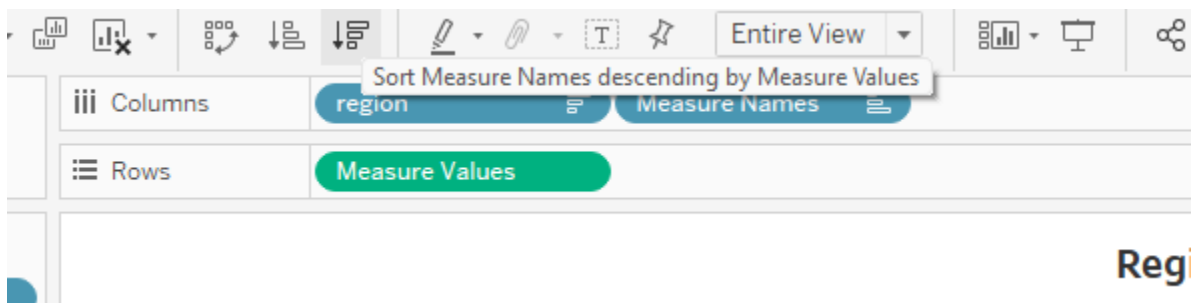
1. Drag the contract type to Columns.
2. Drag Region to Columns.
3. Drag customer_id (count) to Rows.
4. Drag churn to Color.
5. Tooltip: add churn rate, customer count.
6. Add the filters as shown below

7. Format the title and change the view to Entire view



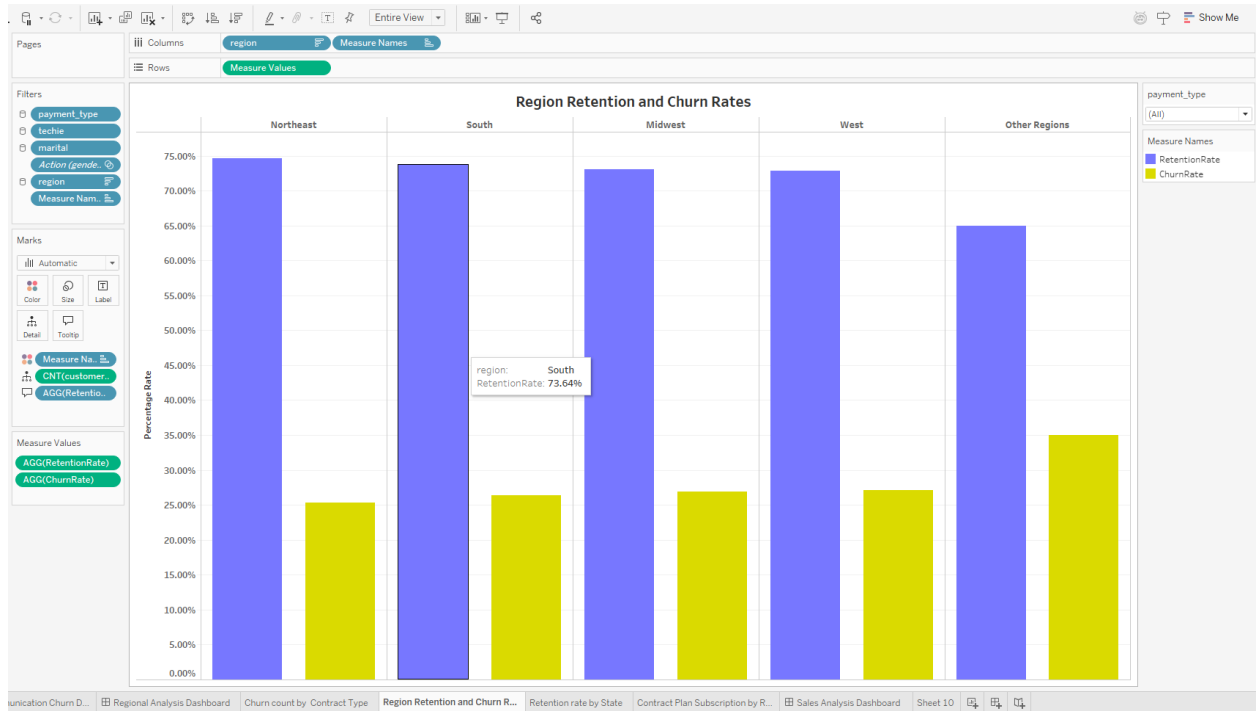
Worksheet 7: Side-by-Side Bar – Retention and Churn Rate

1. Drag the Region to the canvas
2. Drag the Retention rate and churn rate to the canvas
3. Click on the Show Me tab on the right-hand side and choose the side-by-side bar chart visual
4. Format the bars to descend based on the retention rate



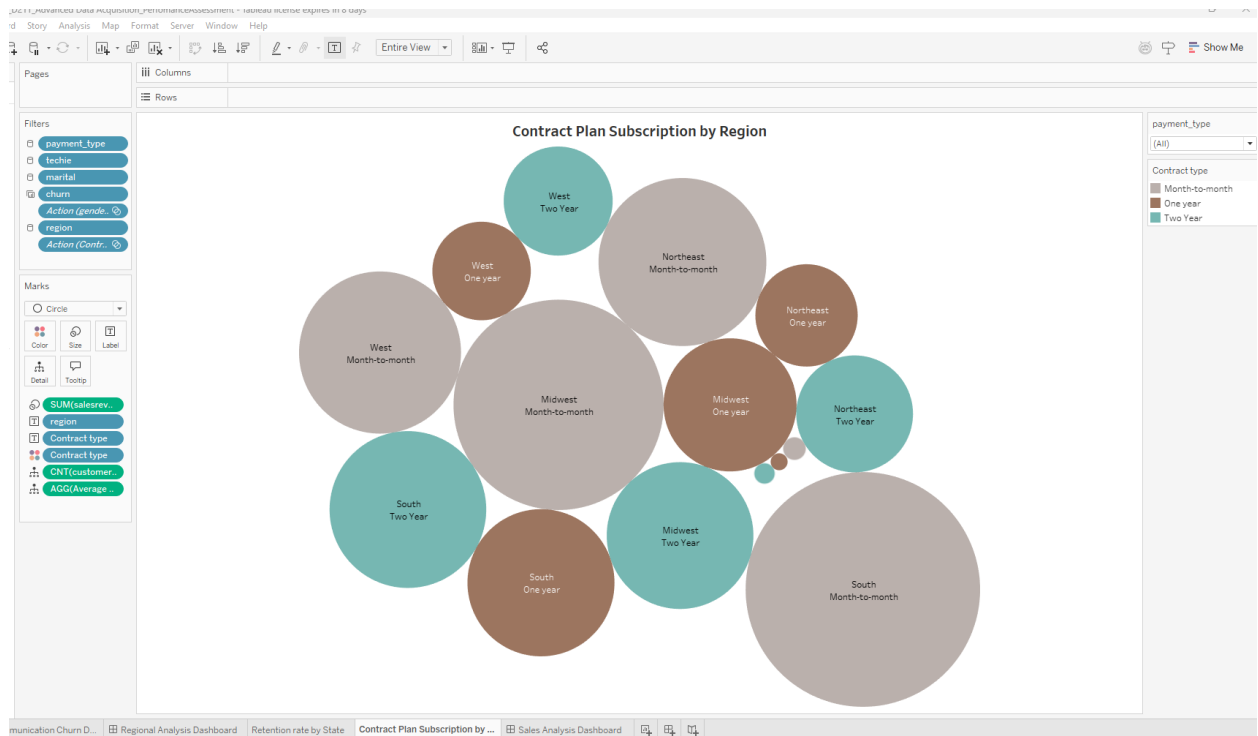
5. Add the filters as shown below

6. Format the title and change the view to Entire view



Worksheet 8: Bubble Chart – Contract Subscription by Region

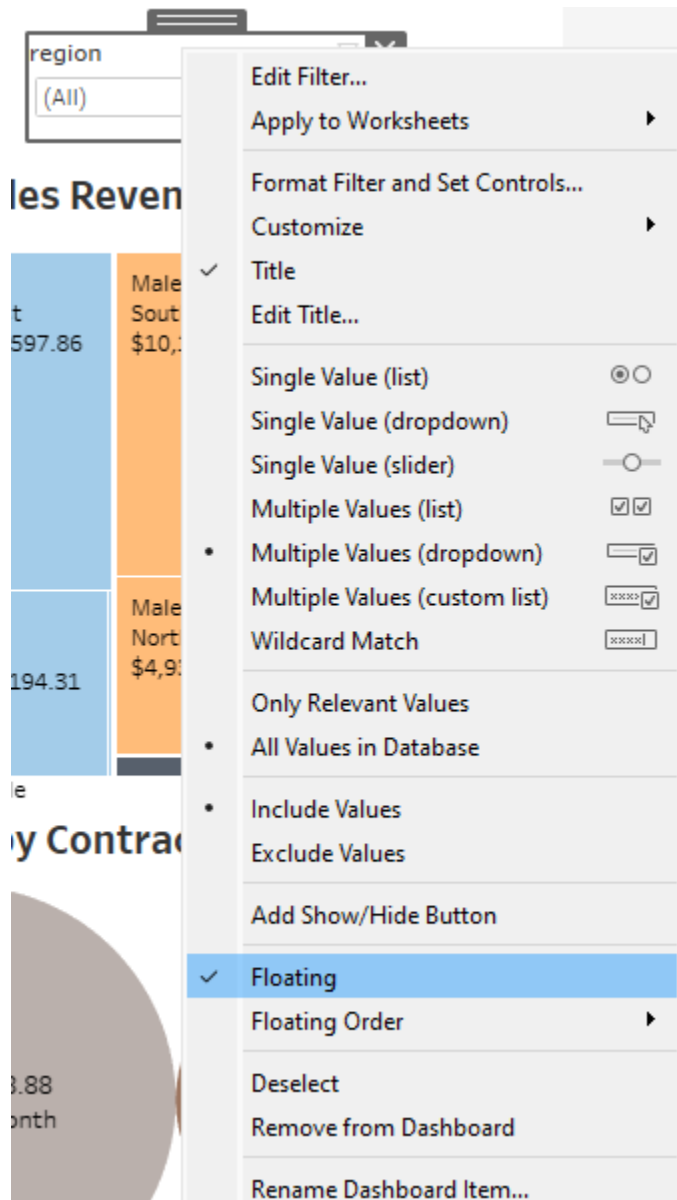
1. Drag the Contract to the labels on the Marks
2. Drag the region to size on the Marks
3. Click on the Show Me tab on the right-hand side and choose the bubble chart visual
4. Drag the Contract Type to the color.
5. Drag SalesRevenue to Size.
6. Drag average monthly_charge to Tooltip and customer_id (count) to Tooltip.
7. Add the filters as shown below
8. Format the title and change the view to Entire view



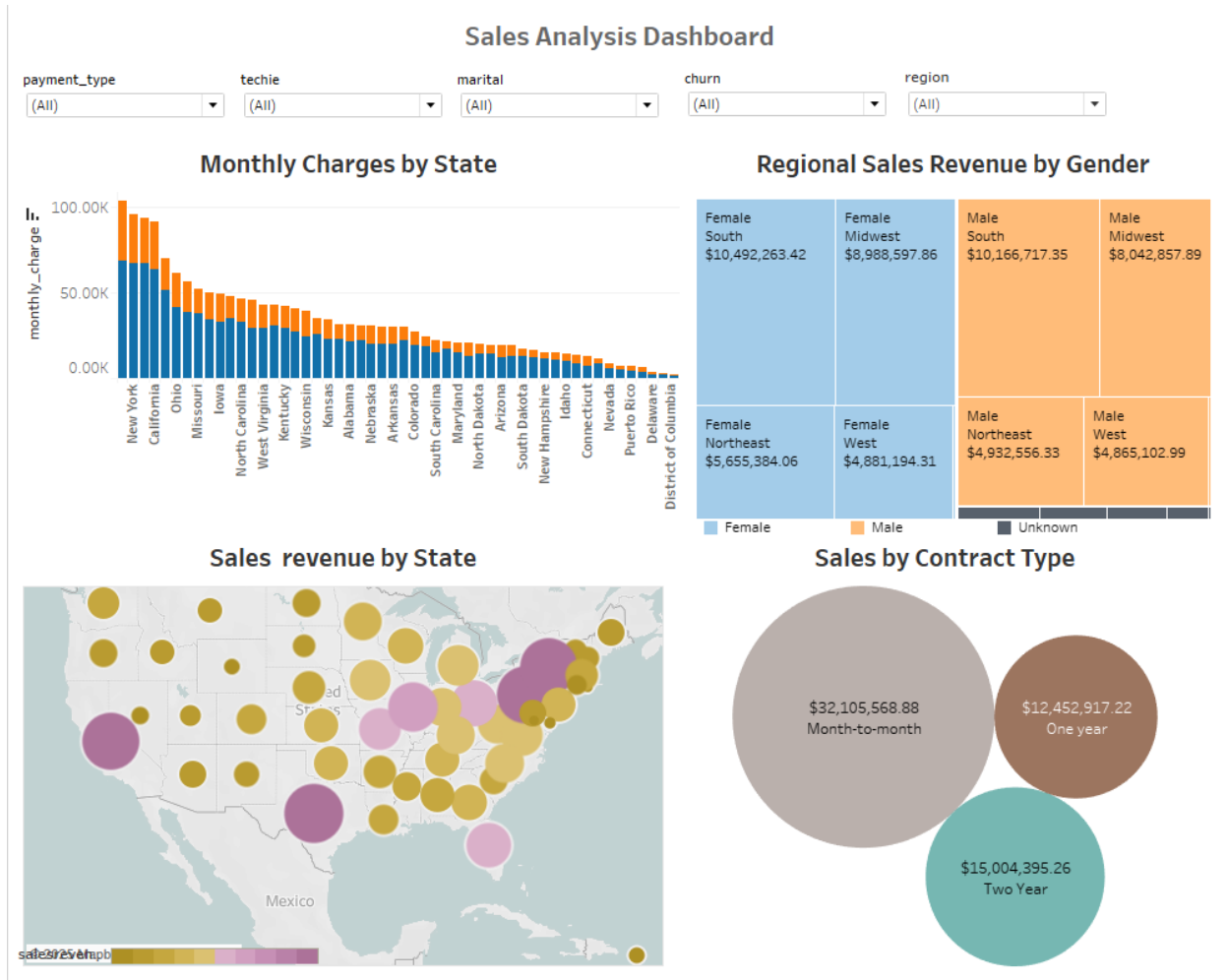
Step 5: Creation of the dashboards and story

• Dashboard 1: Sales Analysis Dashboard

1. Click on the Dashboard tab > New Dashboard.
2. Name it: Sales Analysis Dashboard.
3. Drag and arrange the following worksheets:
 - a. Stacked Bar (Monthly Charges by State)
 - b. Heatmap (Regional Sales Revenue by Gender)
 - c. Map (Sales revenue by state)
 - d. Bubble Chart (sales by contract type)
4. Make the filters float and move them to the top. Filters on the dashboard include
Payment_type, Techie, Marital, Churn, and Region



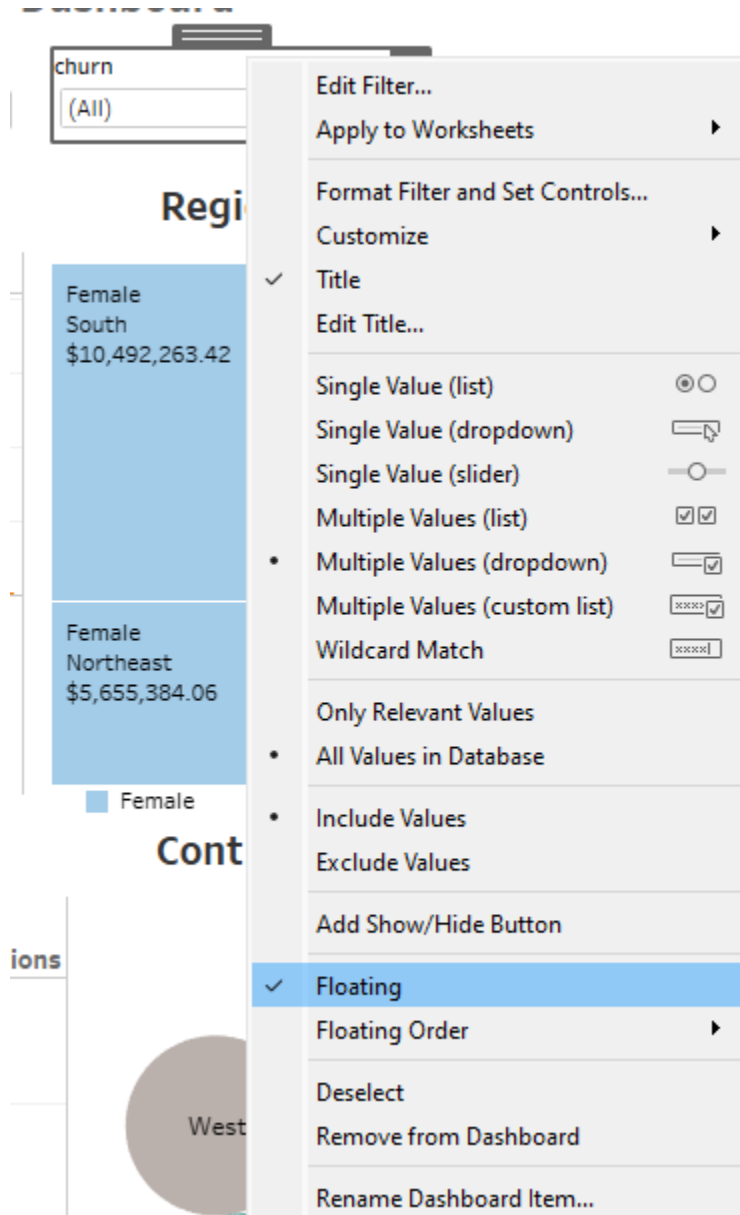
5. Format the title, layout, spacing, and legends.



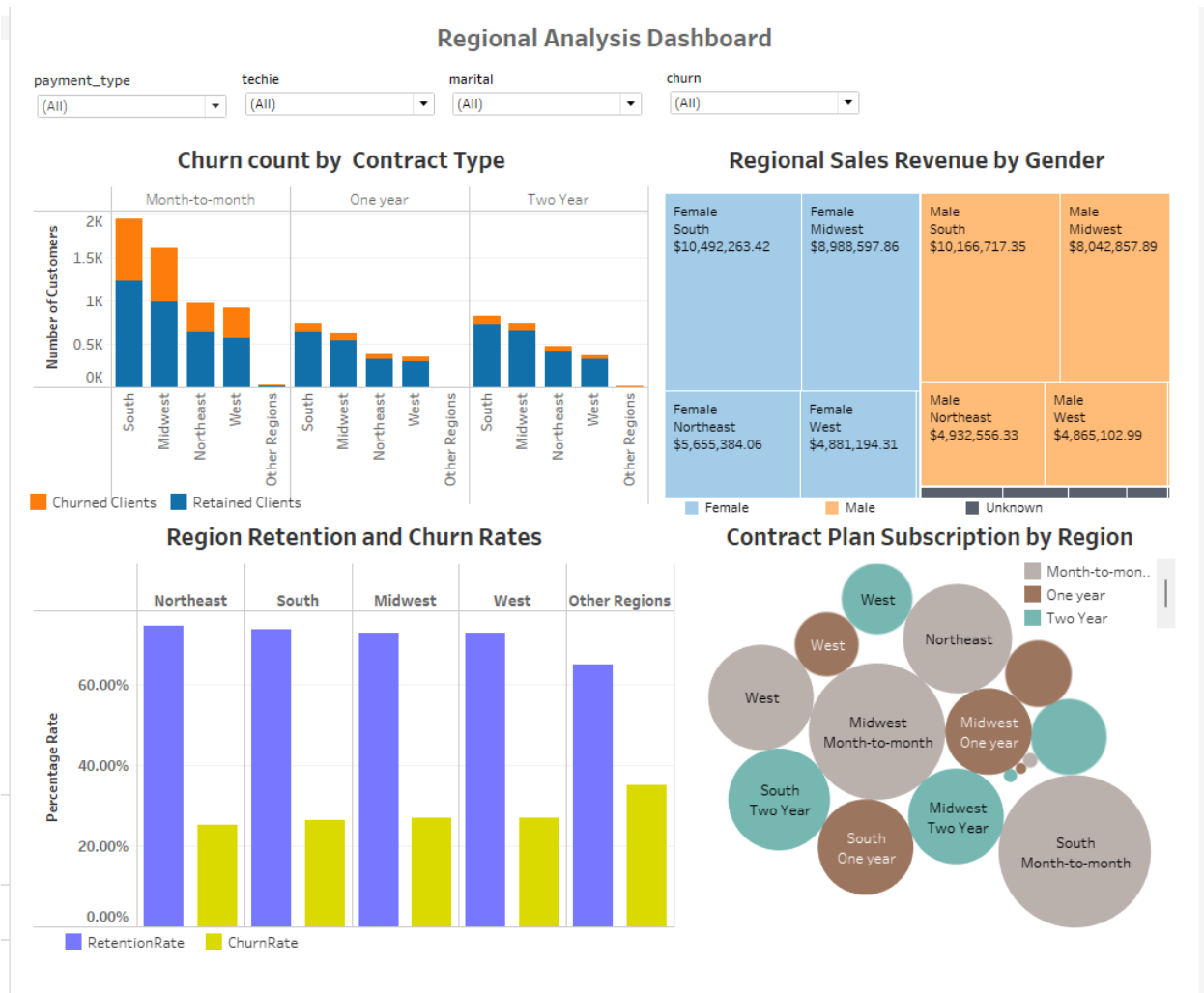
Dashboard 2: Regional Analysis Dashboard

1. Create a new Dashboard.
2. Name it: Regional Analysis Dashboard.
3. Add the following worksheets:
 - a. Stacked Bar (Churn count by Contract type)
 - b. Side-by-Side Bar (Region Retention and churn rates)
 - c. Heatmap (Regional Sales Revenue by Gender)
 - d. Bubble Chart (Contract Plan Subscription by Region)

4. Make the filters float and move them to the top. Filters on the dashboard include
Payment_type, Techie, Marital and Churn



5. Format the title, layout, spacing, and legends.



Adding the dashboards to the Telecommunication Churn Data Analysis Story

1. Click on the Story tab > New Story.
2. Name it: Telecommunication Churn Data Analysis
3. Add two story points:
 - a. Page 1: Regional Analysis, name the caption Regional Analysis Dashboard.
Insert Regional Analysis Dashboard.
 - b. Page 2: Sales Analysis, name the caption Sales Analysis Dashboard.
Insert Sales Analysis Dashboard.

Telecommunication Churn Data Analysis



Regional Analysis Dashboard

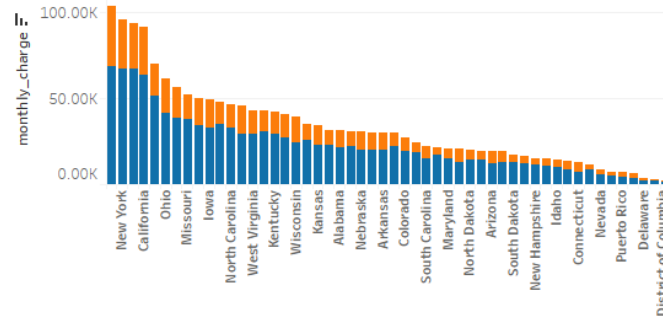
Sales Analysis Dashboard



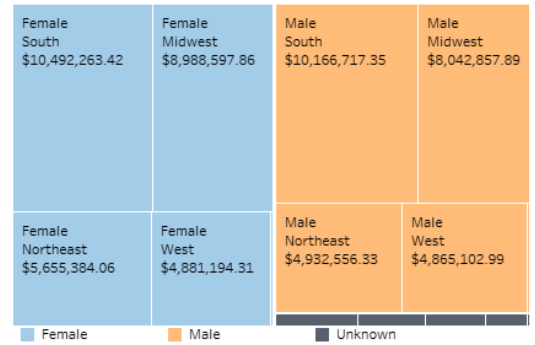
Sales Analysis Dashboard

payment_type (All) techie (All) marital (All) churn (All) region (All)

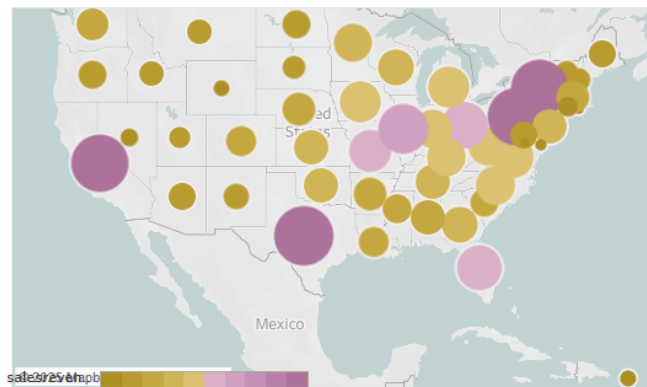
Monthly Charges by State



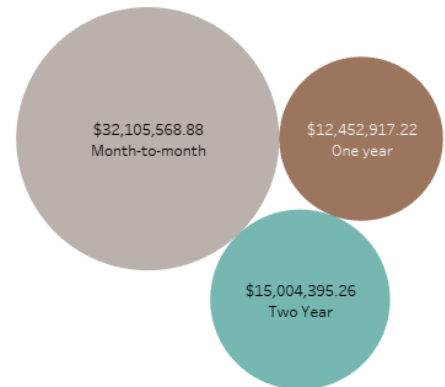
Regional Sales Revenue by Gender



Sales revenue by State



Sales by Contract Type



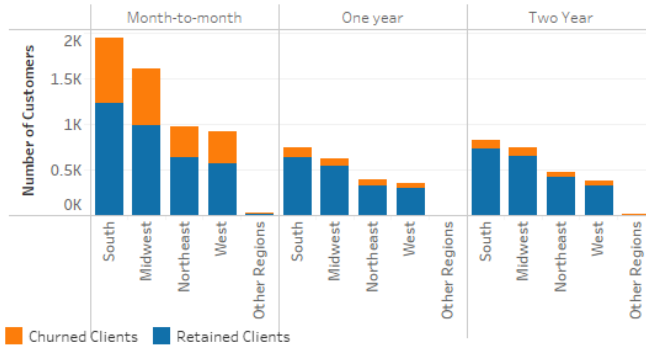
Telecommunication Churn Data Analysis

< Regional Analysis Dashboard Sales Analysis Dashboard >

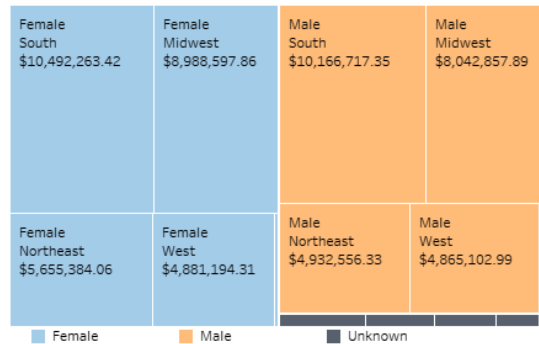
Regional Analysis Dashboard

payment_type (All) techie (All) marital (All) churn (All)

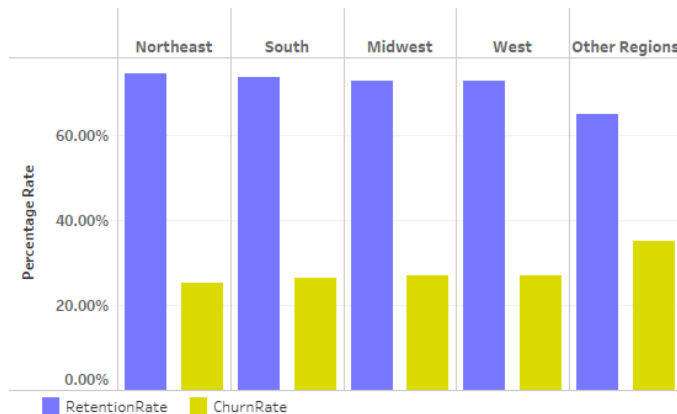
Churn count by Contract Type



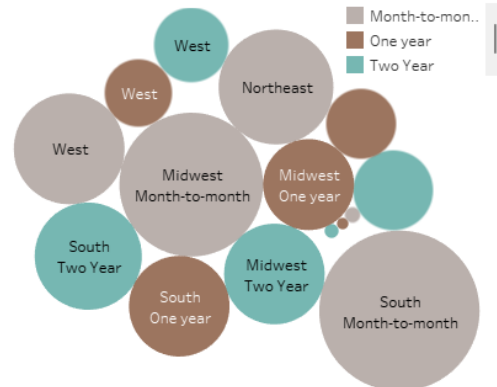
Regional Sales Revenue by Gender



Region Retention and Churn Rates



Contract Plan Subscription by Region



C5: DATA ANALYSIS RESULTS

The Telecommunications Churn data analysis story has two dashboards, comprising the Sales Analysis Dashboard and the Regional Analysis Dashboard, to provide executive leadership in the telecommunications industry with actionable insights into customer behavior, sales performance, contract trends, and regional dynamics. These dashboards collectively support strategic decision-making around customer retention, revenue growth, and operational focus.

The Sales Analysis Dashboard was designed to support executive decision-making by providing insights into customer behavior, sales revenue trends, and demographic patterns that affect churn and retention. The dashboard incorporates visualizations that allow stakeholders to interact with the data, spot regional variations, and make informed strategic decisions based on contract types, gender, geographic distribution, and customer value.

1. Monthly Charges and Churn Insights by State

The stacked bar chart revealed that Texas, New York, California, Ohio, and Missouri are the top contributors to overall monthly charges, regardless of churn status. This finding aligns with the dashboard's goal of identifying high-revenue states and analyzing how churn impacts revenue across locations. The ability to distinguish between retained and churned customers directly supports efforts to refine retention strategies in high-value markets.

2. Gender-Based Revenue Trends Across Regions

The heatmap titled "Regional Sales Revenue by Gender" highlighted a notable revenue concentration among female customers in the South region, accounting for \$10,492,263.42. This demographic insight reinforces the purpose of the dashboard by exposing key customer segments that drive revenue. The top five contributing states—Texas, Virginia, Florida, Kentucky, and North Carolina—indicate where targeted marketing or customer engagement strategies could be most effective for this demographic. Furthermore, regional monthly charge patterns spotlight Texas, Virginia, West Virginia, and Georgia as recurring high-value areas, further reinforcing their strategic importance.

3. Total Sales Revenue by State

Using the map visualization, Texas emerged as the top-performing state with a total sales revenue of \$3,496,988.77. Filtering down to gender and contract type, the analysis showed that:

Female customers contributed \$1,795,608.58

Month-to-month contracts were responsible for \$2,032,117.56 in revenue

Retained customers in Texas generated \$69,000 in monthly charges with an average monthly fee of \$161.85

These insights reinforce how different segments contribute to revenue, validating the dashboard's function in uncovering opportunities for both customer retention and contract optimization.

4. Contract Subscription Revenue Patterns

The bubble chart, "Sales by Contract Type," revealed that Month-to-Month contracts dominate revenue generation, contributing \$32,105,568.88. Within this group, the top states for sales revenue over customer tenure were Texas, Pennsylvania, California, New York, and Illinois, while New York, Pennsylvania, Florida, Missouri, and Iowa led in monthly charges. The dashboard further illustrated that females in the South region are the top contributors, followed by males in the same region, highlighting a consistent regional and gender-based revenue pattern.

The Regional Analysis Dashboard was developed to offer stakeholders a deeper understanding of customer retention patterns and sales performance across geographic regions. Its purpose is to inform executive decisions related to regional marketing strategies, customer retention efforts, and contract management. The visualizations provide a comprehensive overview of how customer behavior varies by region, gender, and contract type, enabling targeted and data-driven actions.

1. Regional Customer Distribution and Churn by Contract Type

The Stacked Bar Chart revealed that the South region consistently has the largest customer base across all contract types, followed by the Midwest, Northeast, West, and other regions. However,

across every region, the Month-to-Month contract type exhibited the highest churn rates, indicating instability and a lack of long-term customer commitment. This finding supports the dashboard's function by highlighting where customer loyalty is weakest, providing a focal point for intervention. Executives can use this insight to consider strategies like improved service offerings or discounts for long-term contracts in high-churn areas.

2. Retention vs. Churn Rates Across Regions

The Side-by-Side Bar Chart comparing churn and retention rates per region showed that the Northeast region has the highest retention rate at 74.69%, whereas other regions have the lowest at 65%.

This comparison directly supports the purpose of the dashboard by identifying regions with strong customer loyalty and those needing retention-focused initiatives. Stakeholders can explore what contributes to higher loyalty in the Northeast and replicate similar strategies in weaker-performing regions.

3. Regional Sales Revenue by Gender

The Heatmap, titled Regional Sales Revenue by Gender, emphasized that the highest total sales revenue of \$10,492,263.42 was generated by female customers in the South region.

Top contributing states include Texas, Virginia, Florida, Kentucky, and North Carolina. These insights support gender- and region-specific marketing strategies and align with the dashboard's function to expose high-value customer segments.

Additionally, the top states for monthly charges are Texas, Virginia, West Virginia, and Georgia, which serve as indicators of high customer spending and opportunities for revenue protection through retention efforts.

4. Contract Subscription Trends by Region

The Bubble Chart visualization showed that the South region's Month-to-Month contract type contributed the highest sales revenue at \$11,458,195.57, from 1,942 customers with an average monthly charge of \$171.89.

When this datapoint is filtered, deeper patterns emerge:

- Retention Rate: 63.13%
- Churn Rate: 36.87%
- Customers Retained: 1,226
- Customers Churned: 716
- Top Gender Contributor: Female customers generated \$5,622,225.06
- Top Revenue-Contributing States: Texas, Florida, Virginia, West Virginia, Georgia

These insights confirm that although the South drives significant revenue, it also experiences high churn, especially under the Month-to-Month contract. This dual insight aligns directly with the dashboard's purpose by helping decision-makers balance customer acquisition with retention.

The telecommunication churn data analysis story depicted by the Sales Analysis and Regional Analysis dashboards transforms raw data into strategic insights that align with executive-level goals, including:

- Enhancing customer retention strategies
- Identifying high-value segments by region, contract, and demographic
- Optimizing revenue by focusing on profitable contract types and locations
- Reducing churn by targeting at-risk customer groups
- Supporting data-driven decisions through dynamic filters and visual storytelling

These dashboards serve as a unified platform for understanding customer patterns and improving business outcomes in a competitive telecommunications environment. The dashboards can also

be filtered by key customer characteristics such as payment type, marital status, tech support availability, and churn status. These dimensions further enhance the dashboard's functionality by allowing stakeholders to explore how various customer attributes influence revenue, engagement, and loyalty, making it a truly comprehensive decision-support tool.

C6: ANALYSIS LIMITATIONS

The analysis was limited due to a lack of understanding of the business data, how the data was collected, and the domain knowledge. Moreover, the data lacked appropriate dates to enable analysis and trends of the effect of various factors over a time period.

D : WEB SOURCES

Shacham Omer (2018) USA States to Region (States.csv)

<https://www.kaggle.com/datasets/omer2040/usa-states-to-region/data>

E : SOURCES

F : PROFESSIONAL COMMUNICATION